Carl Friedrich Gauss

Works, Volume IV

Theory of the Combination of Observations Subject to the Smallest Errors

First Part and Second Part

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Related English mathematical-argument usage was consulted in James Caddall Morehead and Adam Miller Hiltebeitel's *General Investigations of Curved Surfaces of 1827 and 1825* (1902), Project Gutenberg 36856, TeX edition. Related German quantity and mathematical-argument usage was consulted in H. Maser's translation and edition, *Carl Friedrich Gauss' Untersuchungen über höhere Arithmetik* (1889), historical digitized edition. These works are not parallel translations of the observation theory. Gauss's German notice of 26 February 1821, in the same historical volume, was consulted for related error and probability terminology. No human running prose or formulas are adopted from these references. Historical publication titles remain in the original. Exact versions and URLs are retained in `CITATIONS.bib`.

For the Second Part, related English expression, substitution and mathematical-argument usage in Morehead and Hiltebeitel and related German quantity, substitution and equation usage in Maser were consulted. These are not parallel translations, attestations of the historical wording or authority for the formulas or proof. No human running prose or modern formulas are adopted. The German notice of 1821 was consulted for the First Part only. The Second Part was presented on 2 February 1823 and published in 1823.

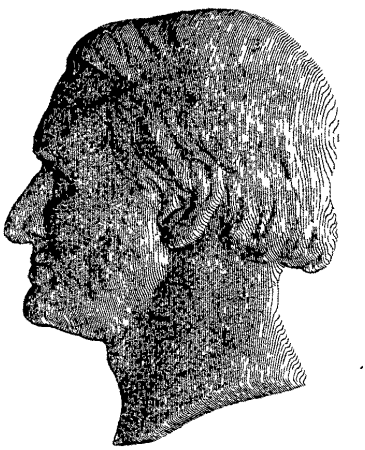
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THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS

FIRST PART

BY

CARL FRIEDRICH GAUSS

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THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS.

First Part.

1.

However much care is taken in making observations concerned with the magnitudes of natural things, they nevertheless always remain subject to larger or smaller errors. Errors of observation are usually not simple, but arise from several sources at once: two kinds of these sources must be carefully distinguished. Some causes of error are such that their effect in any observation depends on variable circumstances between which and the observation itself no essential connection is understood: the errors arising from these causes are called irregular or random, and insofar as those circumstances cannot be subjected to calculation, the same holds for the errors themselves. Such are the errors arising from the imperfection of the senses, and also from irregular external causes, e.g. the quivering motion of the air, which slightly disturbs vision: several defects of even the best instruments must also be included here, e.g. roughness of the inside of spirit levels, lack of absolute stability, etc. Other causes of error, by contrast, by their nature produce in all observations of the same kind an effect that is either absolutely constant, or at least one whose magnitude depends, according to a definite law, solely on circumstances regarded as essentially connected with the observation. Errors of this kind are called constant or regular.

It is evident, moreover, that this distinction is relative to some extent, and depends on the wider or narrower sense in which the notion of observations belonging to the same kind is taken. For example, irregular defects in the graduation of instruments for measuring angles produce a constant error whenever the question is merely one of repeating the observation of a particular angle indefinitely, since the same faulty divisions are always used in this case: on the other hand, the error arising from this source can be regarded as random whenever the question is indefinitely one of measuring angles of any magnitude, provided there is no table showing the amount of error in the individual divisions.

2.

Strictly speaking, the consideration of regular errors is excluded from our purpose. It is the observer's task to investigate diligently all causes capable of producing constant errors, and either to remove them, or at least to examine

their nature and magnitude with the utmost care, so that their effect can be determined in any particular observation. The observation can thus be freed from that effect, which amounts to the same thing as if the error had not been present at all. The situation is very different, however, with irregular errors, which by their nature cannot be subjected to calculation. These must therefore be tolerated in the observations, but their effect on the quantities to be derived from the observations must be reduced as much as possible by a skilful combination of the latter. The following investigations are devoted to this most important subject.

3.

Errors in observations belonging to the same kind that arise from a particular simple cause are, by the nature of the matter, confined within certain *limits*, which could undoubtedly be assigned exactly if the nature of the cause itself were *fully* understood. Most causes of random errors are such that, according to the law of continuity, all errors contained within those limits must be regarded as possible, and perfect knowledge of the cause would also teach us whether all these errors are equally likely or not equally likely. In the latter case it would also teach us what relative probability is to be assigned to each error. The same will hold for the total error, compounded from several simple errors: namely, it will be enclosed within certain limits, (one of which will be equal to the sum of all the partial upper limits, the other to the sum of all the lower limits); all errors within these limits will indeed be possible, but since each can be composed in infinitely many different ways from partial errors that are themselves more or less probable, we shall have to assign a greater chance to one and a smaller chance to another, and the law of relative probability can be found if the laws of the simple errors are assumed known, subject to the analytical difficulties of collecting all the combinations.

There are, indeed, some causes of error that can produce only discrete errors, rather than errors proceeding according to the law of continuity, such as errors in the graduation of instruments, (if one chooses to count these among random errors): for the number of divisions in any particular instrument is finite. It is evident, however, that notwithstanding this, provided that not all the causes of error produce discrete errors, the collection of all possible total errors will constitute a series proceeding according to the law of continuity, or several such interrupted series if, when all possible discrete errors are arranged in order of magnitude, one or another difference between two adjacent terms happens to be greater than the difference between the limits of the total errors insofar as they arise from continuous errors alone. But in practice the latter case will hardly ever occur, unless the graduation suffers from rather gross defects.

4.

If we denote the relative likelihood of the total error x, in a particular kind of observation, by the characteristic φx, then, because of the continuity of the errors, this is to be understood as meaning that the probability of an error between the infinitely close limits x and $x + \mathrm{d}x$ is $= \varphi x.\mathrm{d}x$. In practice it will hardly ever, indeed scarcely ever at all, be possible to assign this function a priori: nevertheless several general results concerning it can be established, which we shall present below. The following is evident. The function φx must be classed among discontinuous functions insofar as, for all values of x lying outside the limits of the possible errors, it must be $= 0$; within these limits, however, it will take a positive value everywhere (leaving aside the case of which we spoke at the end of the preceding article). In most cases it will be permissible to suppose positive and negative errors of the same magnitude equally likely, in which case $\varphi(-x) = \varphi x$. Further, since smaller errors are more readily committed than larger ones, the value of φx will usually be greatest for $x = 0$, and will decrease continuously as x increases.

In general, however, the value of the integral $\int \varphi x.\mathrm{d}x$, extended from $x = a$ to $x = b$, will express the probability that an error not yet known lies between the limits a and b. The value of this integral from the lower limit of all possible errors to the upper limit will therefore always be $= 1$. And since φx is, for all values of x outside these limits, always $= 0$, it is evident also that

the value of the integral $\int \varphi x.\mathrm{d}x$ *extended from* $x = -\infty$ *to* $x = +\infty$ *is always* $= 1$.

5.

Let us further consider the integral $\int x\varphi x.\mathrm{d}x$ between the same limits, whose value we shall put $= k$. If all the simple causes of error are such that there is no reason why one of two errors equal in magnitude but opposite in sign should be produced more readily than the other, this will also hold for the total error, or we shall have $\varphi(-x) = \varphi x$, and consequently, necessarily, $k = 0$. From this we conclude that whenever k does not vanish, but is, for example, a positive quantity, there must necessarily be one or another cause of error that can either produce positive errors only, or at least produce positive errors more readily than negative ones. This quantity k, which is in fact the mean of all possible errors, or the mean value of x, can conveniently be called the constant part of the error. Moreover, it can easily be proved that the constant part of the total error is equal to the sum of the constant parts contained in the errors arising from the individual simple causes. If the quantity k is supposed known and is subtracted from every observation, and if the error of the observation thus corrected is denoted by x', and its probability by $\varphi' x'$, we shall have $x' = x - k$, $\varphi' x' = \varphi x$ and consequently $\int x'\varphi' x'.\mathrm{d}x' = \int x\varphi x.\mathrm{d}x - \int k\varphi x.\mathrm{d}x = k - k = 0$, i.e. the errors of the

corrected observations will have no constant part, which is also evident in itself.

6.

Just as the integral $\int x\varphi x.\mathrm{d}x$, or the mean value of x, tells us either the absence of a constant error or its presence and magnitude, the integral

$$\int xx\varphi x.\mathrm{d}x$$

extended from $x = -\infty$ to $x = +\infty$ (or the mean value of the square xx) seems best suited to defining and measuring the uncertainty of observations in general, so that, of two systems of observations differing in the likelihood of their errors, those in which the integral $\int xx\varphi x.\mathrm{d}x$ has the smaller value are to be judged superior in precision. If anyone objects that this way has been chosen arbitrarily, without any compelling necessity, we shall readily agree. For this question, by its nature, involves something indefinite that cannot be confined within bounds except by a principle that is to some extent arbitrary. The determination of a quantity by an observation subject to a larger or smaller error may aptly be compared to a game in which there are losses only and no gains, since every error to be feared is akin to a loss. The cost of such a game is assessed by the probable loss, namely by the sum of the products of the individual possible losses and their respective probabilities. But what loss should be taken as equivalent to each error of observation is by no means evident in itself; rather, this determination depends to some extent on our choice. It is evidently not permissible to put the loss equal to the error itself; for if positive errors were taken as losses, negative ones would have to represent gains. The magnitude of the loss must instead be expressed by a function of the error that by its nature is always positive. Since the variety of such functions is infinite, the simplest that has this property seems to be preferred to the others, and this is indisputably the square: in this way the principle stated above emerges.

The illustrious Laplace did indeed consider the matter in a similar way, but adopted the error itself, always taken positively, as the measure of loss. But unless we are mistaken, this way is at least no less arbitrary than ours: for whether a double error is considered as tolerable as a single one repeated twice, or less tolerable, and consequently whether it is more appropriate to assign to a double error a weight only twice as great or a greater one, is a question neither evident in itself nor to be decided by mathematical demonstrations, but one to be left solely to free choice. Furthermore, it cannot be denied that continuity is violated in that way: and for this very reason that method is less amenable to analytical treatment, whereas the results to which our principle leads commend themselves by their remarkable simplicity as well as generality.

7.

Putting the value of the integral $\int xx\varphi x.\mathrm{d}x$ extended from $x = -\infty$ to $x = +\infty = mm$, we shall call the quantity m the *mean error to be feared*, or simply the *mean error*, of observations whose indeterminate errors x have the relative probability φx. We shall not restrict this name to direct observations, but shall also extend it to any determinations derived from observations. Care must be taken, however, not to confuse the mean error with the arithmetic mean of all the errors, of which we spoke in art. 5.

Suppose several kinds of observations, or several determinations obtained from observations, to which the same precision cannot be assigned, are being compared. Their relative *weight* for us will be a quantity inversely proportional to mm, while their *precision* is taken as simply inversely proportional to m. In order, therefore, that a weight may be expressed by a number, the weight of a particular kind of observation must have been taken as the unit.

8.

If the errors of observations contain a constant part, removing it diminishes the mean error and increases the weight and precision. Retaining the notation of art. 5, and denoting the mean error of the corrected observations by m', we shall have

$$\begin{aligned} m'm' &= \int x'x'\varphi'x'.\mathrm{d}x' = \int (x-k)^2\varphi x.\mathrm{d}x \\ &= \int xx\varphi x.\mathrm{d}x - 2k\int x\varphi x.\mathrm{d}x + kk\int \varphi x.\mathrm{d}x \\ &= mm - 2kk + kk = mm - kk. \end{aligned}$$

If, however, instead of the true constant part k, another quantity l had been subtracted from the observations, the square of the new mean error would become $= mm - 2kl + ll = m'm' + (l-k)^2$.

9.

If λ denotes a definite coefficient, and μ the value of the integral $\int \varphi x.\mathrm{d}x$ from $x = -\lambda m$ to $x = +\lambda m$, then μ will be the probability that the error of an observation is less than λm (without regard to sign), and $1-\mu$ the probability of an error greater than λm. If, therefore, the value $\mu = \frac{1}{2}$ corresponds to the value $\lambda m = \rho$, the error can fall below ρ just as readily as above ρ, so that ρ can conveniently be called the *probable* error. The relation between the quantities λ, μ evidently depends on the nature of the function φx, which is usually unknown. It will therefore be worthwhile to consider this relation more closely in certain special cases.

I. If the limits of all possible errors are $-a$ and $+a$, and all errors within these limits are equally probable, φx will be constant between the limits $x = -a$ and $x = +a$, and consequently $= \frac{1}{2a}$. Hence $m = a\sqrt{\frac{1}{3}}$, and also

$\mu = \lambda\sqrt{\frac{1}{3}}$, as long as λ is not greater than $\sqrt{3}$; finally, $\rho = m\sqrt{\frac{3}{4}} = 0{,}8660254m$, and the probability that an error no greater than the mean error occurs will be $= \sqrt{\frac{1}{3}} = 0{,}5773503$.

II. If, as before, $-a$ and $+a$ are the limits of the possible errors, and the probability of the errors themselves is supposed to decrease on both sides from the error 0 in arithmetic progression, we shall have

$$\varphi x = \frac{a - x}{aa}, \quad \text{for values of } x \text{ between 0 and } +a$$
$$\varphi x = \frac{a + x}{aa}, \quad \text{for values of } x \text{ between 0 and } -a$$

From this we deduce $m = a\sqrt{\frac{1}{6}}$, $\mu = \lambda\sqrt{\frac{2}{3}} - \frac{1}{6}\lambda\lambda$, as long as λ is between 0 and $\sqrt{6}$; finally, $\lambda = \sqrt{6} - \sqrt{(6 - 6\mu)}$, as long as μ is between 0 and 1, and consequently

$$\rho = m(\sqrt{6} - \sqrt{3}) = 0{,}7174389m$$

The probability of an error not exceeding the mean error will in this case be

$$= \sqrt{\frac{3}{2}} - \frac{1}{6} = 0{,}6498299$$

III. If we put the function φx proportional to $e^{-\frac{xx}{hh}}$ (which can indeed be only approximately true in nature), we must have

$$\varphi x = \frac{e^{-\frac{xx}{hh}}}{h\sqrt{\pi}}$$

where π denotes the semicircumference of a circle of radius 1, whence we further deduce

$$m = h\sqrt{\frac{1}{2}}$$

(See *Disquis. generales circa seriem infinitam* etc. art. 28). Further, if the value of the integral

$$\frac{2}{\sqrt{\pi}} \int e^{-zz} \mathrm{d}z$$

beginning at $z = 0$ is denoted by Θz, we shall have

$$\mu = \Theta(\lambda\sqrt{\tfrac{1}{2}})$$

The following table gives some values of this quantity:

λ	μ
0,6744897	0,5
0,8416213	0,6
1,0000000	0,68268·95
1,0364334	0,7
1,2815517	0,8
1,6448537	0,9
2,5758293	0,99
3,2918301	0,999
3,8905940	0,9999
∞	1

10.

Although the relation between λ and μ depends on the nature of the function φx, some general results can nevertheless be established. Namely, whatever this function may be, provided it is such that its value always decreases, or at least does not increase, as the absolute value of x increases, we shall certainly have

> λ less than, or at least not greater than, $\mu\sqrt{3}$, whenever μ is less than $\frac{2}{3}$;
>
> λ not greater than $\frac{2}{3\sqrt{(1-\mu)}}$, whenever μ is greater than $\frac{2}{3}$.

For $\mu = \frac{2}{3}$ the two limits coincide: namely, λ cannot be greater than $\sqrt{\frac{4}{3}}$.

To prove this remarkable theorem, let us denote by y the value of the integral $\int \varphi z.\mathrm{d}z$ extended from $z = -x$ to $z = +x$, so that y will be the probability that an error is contained within the limits $-x$ and $+x$. Further, let us put

$$x = \psi y, \quad \mathrm{d}\psi y = \psi' y.\mathrm{d}y, \quad \mathrm{d}\psi' y = \psi'' y.\mathrm{d}y$$

We shall therefore have $\psi 0 = 0$, and also

$$\psi' y = \frac{1}{\varphi x + \varphi(-x)}$$

so that, by hypothesis, $\psi' y$ will always increase from $y = 0$ to $y = 1$, or at least will nowhere decrease; or, what is the same, the value of $\psi'' y$ will always be positive, or at least not negative. Further, we have $\mathrm{d}.y\psi' y = \psi' y.\mathrm{d}y + y\psi'' y.\mathrm{d}y$, and consequently

$$y\psi' y - \psi y = \int y\psi'' y.\mathrm{d}y$$

the integration beginning at $y = 0$. The value of the expression $y\psi' y - \psi y$ will therefore always be a positive quantity, or at least not a negative one, and consequently

$$1 - \frac{\psi y}{y\psi' y}$$

will be a positive quantity less than unity. Let f be its value for $y = \mu$, i.e. since we have $\psi\mu = \lambda m$, let

$$f = 1 - \frac{\lambda m}{\mu\psi'\mu} \quad \text{or} \quad \psi'\mu = \frac{\lambda m}{(1-f)\mu}$$

With these preparations, let us consider the following function of y

$$\frac{\lambda m}{(1-f)\mu}(y - \mu f)$$

which we shall put $= Fy$, and also $\mathrm{d}Fy = F'y.\mathrm{d}y$. It is evident that

$$F\mu = \lambda m = \psi\mu$$

$$F'\mu = \frac{\lambda m}{(1-f)\mu} = \psi'\mu$$

Since, therefore, $\psi'y$ increases continuously as y increases (or at least does not decrease, which is always to be understood), while $F'y$ is constant, the difference $\psi'y - F'y = \frac{\mathrm{d}(\psi y - Fy)}{\mathrm{d}y}$ will be positive for values of y greater than μ, and negative for smaller ones. From this it is easily concluded that $\psi y - Fy$ is always a positive quantity, and consequently ψy will always be absolutely greater, or at least not smaller, than Fy, certainly as long as the value of Fy is positive, i.e. from $y = \mu f$ to $y = 1$. Hence the value of the integral $\int (Fy)^2 \mathrm{d}y$ from $y = \mu f$ to $y = 1$ will be smaller than the value of the integral $\int (\psi y)^2 \mathrm{d}y$ between the same limits, and therefore still more certainly smaller than the value of this integral from $y = 0$ to $y = 1$, which is $= mm$. But the value of the former integral is found to be

$$= \frac{\lambda\lambda mm(1-\mu f)^3}{3\mu\mu(1-f)^2}$$

whence we conclude that $\lambda\lambda$ is less than $\frac{3\mu\mu(1-f)^2}{(1-\mu f)^3}$, where f is a quantity lying between 0 and 1. Now the value of the fraction $\frac{3\mu\mu(1-f)^2}{(1-\mu f)^3}$, whose differential, if f is regarded as a variable quantity, is $=$

$$-\frac{3\mu\mu(1-f)}{(1-\mu f)^4}.(2-3\mu+\mu f)\mathrm{d}f$$

decreases continuously as f passes from the value 0 to the value 1, whenever μ is less than $\frac{2}{3}$; consequently the greatest possible value will be that corresponding to $f = 0$, namely $= 3\mu\mu$, so that in this case λ is certainly less than or not greater than $\mu\sqrt{3}$. Q. E. P. On the other hand, whenever μ is greater than $\frac{2}{3}$, the value of that fraction will be greatest for $2 - 3\mu + \mu f = 0$, i.e. for $f = 3 - \frac{2}{\mu}$, whence it becomes $= \frac{4}{9(1-\mu)}$, and therefore in this case λ is not greater than $\frac{2}{3\sqrt{(1-\mu)}}$. Q E. S.

Thus, for example, for $\mu = \frac{1}{2}$, λ certainly cannot be greater than $\sqrt{\frac{3}{4}}$, i.e. the probable error cannot exceed the limit $0{,}8660254m$, to which it was found equal in the first example of art. 9. Further, it is easily concluded from our theorem that μ is not less than $\lambda\sqrt{\frac{1}{3}}$ as long as λ is less than $\sqrt{\frac{4}{3}}$; on the other hand, μ is not less than $1 - \frac{4}{9\lambda\lambda}$ for a value of λ greater than $\sqrt{\frac{4}{3}}$.

11.

Since several problems to be treated below are also connected with the value of the integral $\int x^4 \varphi x . \mathrm{d}x$, it will be worthwhile to develop it for certain special cases. We shall denote the value of this integral extended from $x = -\infty$ to $x = +\infty$ by n^4.

I. For $\varphi x = \frac{1}{2a}$, insofar as x is contained between $-a$ and $+a$, we have $n^4 = \frac{1}{5}a^4 = \frac{9}{5}m^4$.

II. In the second case of art. 6, where $\varphi x = \frac{a \mp x}{aa}$, for values of x between 0 and $\pm a$, we obtain $n^4 = \frac{1}{15}a^4 = \frac{12}{5}m^4$.

III. In the third case, where

$$\varphi x = \frac{e^{-\frac{xx}{hh}}}{a\sqrt{\pi}}$$

we find, by what is set out in the memoir cited above, $n^4 = \frac{3}{4}h^4 = 3m^4$.

Moreover, it can be proved that the value of $\frac{n^4}{m^4}$ is certainly not less than $\frac{9}{5}$, provided that the hypothesis of the preceding article holds.

12.

Let x, x', x'' etc. denote indefinitely the mutually independent errors of observations of the same kind, whose relative probabilities are expressed by the prefixed characteristic φ; and let y denote a given rational function of the indeterminates x, x', x'' etc.: the multiple integral (I)

$$\int \varphi x.\varphi x'.\varphi x'' \ldots\ldots \mathrm{d}x.\mathrm{d}x'.\mathrm{d}x'' \ldots\ldots$$

extended over all values of the indeterminates x, x', x'' for which the value of y falls within the given limits 0 and η will express the probability of a value of y lying indefinitely within 0 and η. Evidently this integral will be a function of η, whose differential we shall put $= \psi\eta.\mathrm{d}\eta$, so that the integral itself becomes equal to the integral $\int \psi\eta.\mathrm{d}\eta$ beginning at $\eta = 0$. In this way the characteristic $\psi\eta$ must at the same time be regarded as expressing the relative probability of any value of y. Since x can be regarded as a function of the indeterminates y, x', x'' etc., which we shall put

$$= f(y, x', x'' \ldots.)$$

the integral (I) will become

$$= \int \varphi.f(y, x', x'' \ldots).\frac{\mathrm{d}f(y, x', x'' \ldots.)}{\mathrm{d}y}. \varphi x'.\varphi x'' \ldots. \mathrm{d}y.\mathrm{d}x'.\mathrm{d}x'' \ldots.$$

where y must be extended from $y = 0$ to $y = \eta$, while the remaining indeterminates must range over all values to which a real value of $f(y, x', x'' \ldots.)$ corresponds. From this it follows that

$$\psi y = \int \varphi.f(y, x', x'' \ldots.).\frac{\mathrm{d}f(y, x', x'' \ldots.)}{\mathrm{d}y}. \varphi x'.\varphi x'' \ldots. \mathrm{d}x'.\mathrm{d}x'' \ldots.$$

the integration, in which y must be regarded as constant, extending over all values of the indeterminates x', x'' etc. that give $f(y, x', x'' \ldots.)$ a real value.

13.

To carry out this integration in practice, knowledge of the function φ, which is usually unknown, would be required: indeed, even if this function were known, in most cases the integration would exceed the powers of analysis. This being so, we shall be unable to assign the probability of the individual values of y: but the case will be different if only the mean value of y is wanted, which arises from the integration of $\int y\psi y.\mathrm{d}y$ extended over all values of y

that it can attain. And since, evidently, for all values that y cannot attain, either because of the nature of the function it expresses (e.g. for negative values, if we had $y = xx + x'x' + x''x'' + \text{etc.}$), or because certain limits are set to the errors x, x', x'' etc. themselves, we must put $\psi y = 0$, the result will evidently be the same if that integration is extended over all real values of y, namely from $y = -\infty$ to $y = +\infty$. Now the integral $\int y\psi y.\mathrm{d}y$ taken between definite limits, namely from $y = \eta$ to $y = \eta'$, is equal to the integral

$$\int y\varphi.f(y, x', x''\ldots.)\frac{\mathrm{d}f(y, x', x''\ldots.)}{\mathrm{d}y}. \varphi x'.\varphi x''\ldots.\mathrm{d}y.\mathrm{d}x'.\mathrm{d}x''\ldots.$$

the integration extending from $y = \eta$ to $y = \eta'$ and over all values of the indeterminates x', x'' etc. to which a real value of $f(y, x', x''\ldots.)$ corresponds; or, what is the same, to the value of the integral

$$\int y\varphi x.\varphi x'.\varphi x''\ldots.\mathrm{d}x.\mathrm{d}x'.\mathrm{d}x''\ldots.$$

using in this integration for y its value expressed in terms of x, x', x'' etc., and extending the integration over all values of these indeterminates to which a value of y lying between η and η' corresponds. From this we conclude that the integral $\int y\psi y.\mathrm{d}y$ extended over all values of y, from $y = -\infty$ to $y = +\infty$, is obtained from the integration of

$$\int y\varphi x.\varphi x'.\varphi x''\ldots.\mathrm{d}x.\mathrm{d}x'.\mathrm{d}x''\ldots.$$

extended over all real values of x, x', x'' etc., namely from $x = -\infty$ to $x = +\infty$, from $x' = -\infty$ to $x' = +\infty$ etc.

14.

Thus, when the function y has been reduced to the form of a sum of terms such as

$$Ax^{\alpha}x'^{\beta}x''^{\gamma}\ldots.$$

the value of the integral $\int y\psi y.\mathrm{d}y$ extended over all values of y, or the mean value of y, will be equal to the sum of terms

$$A \times \int x^{\alpha}\varphi x.\mathrm{d}x \times \int x'^{\beta}\varphi x'.\mathrm{d}x' \times \int x''^{\gamma}\varphi x''.\mathrm{d}x''\ldots.$$

where the integrations must be extended from $x = -\infty$ to $x = +\infty$, from $x' = -\infty$ to $x' = +\infty$ etc.; or, what amounts to the same thing, to the sum of the terms that arise when the individual powers $x^{\alpha}, x'^{\beta}, x''^{\gamma}$ etc. are replaced by their mean values. The truth of this very important theorem could also easily have been derived from other considerations.

15.

Let us apply what we set out in the preceding article to the special case in which

$$y = \frac{xx + x'x' + x''x'' + \text{etc.}}{\sigma}$$

where σ denotes the number of terms in the numerator. The mean value of y is immediately found here $= mm$, taking the symbol m in the same sense as above. The actual value of y in a definite case may indeed turn out greater or less than the mean, just as may the actual value of the single term xx: but the probability that the random value of y will not differ appreciably from the mean mm will approach certainty more and more as the number σ increases. To make this clearer, since it is not in our power to determine the probability itself exactly, let us investigate the mean error to be feared when we suppose $y = mm$. Evidently, by the principles of art. 6, this error will be the square root of the mean value of the function

$$\left(\frac{xx + x'x' + x''x'' + \text{etc.}}{\sigma} - mm\right)^2$$

to find which it is enough to observe that the mean value of a term such as $\frac{x^4}{\sigma\sigma}$ is $= \frac{n^4}{\sigma\sigma}$ (using the symbol n in the sense of art. 11), while the mean value of a term such as $\frac{2xxx'x'}{\sigma\sigma}$ becomes $= \frac{2m^4}{\sigma\sigma}$, whence the mean value of that function is very easily deduced

$$= \frac{n^4 - m^4}{\sigma}$$

From this we learn that, if a sufficiently large supply of mutually independent random errors x, x', x'' etc. is available, an approximate value of m can be obtained from them with great certainty by the formula

$$m = \sqrt{\frac{(xx + x'x' + x''x'' + \text{etc.})}{\sigma}}$$

and that the mean error to be feared in this determination, with regard to the square mm, is

$$= \sqrt{\frac{n^4 - m^4}{\sigma}}$$

Moreover, since the latter formula involves the quantity n, if the only aim is to form some idea of the degree of precision of this determination, it will be enough to adopt some hypothesis concerning the function φ. For example, under the third hypothesis of artt. 9, 11, that error becomes $= mm\sqrt{\frac{2}{\sigma}}$. If this is less appealing, an approximate value of n^4 can be obtained from the errors themselves with the aid of the formula

$$\frac{x^4 + x'^4 + x''^4 + \text{etc.}}{\sigma}$$

In general, however, we can assert that doubling the precision in this determination requires a fourfold supply of errors, or that the weight of the determination is proportional to the number σ.

In exactly the same way, if the errors of observations involve a constant part, an approximate value of this part can be inferred from the arithmetic mean of many errors with greater confidence the greater their number. And the mean error to be feared in this determination will be expressed by

$$\sqrt{\frac{mm - kk}{\sigma}}$$

if k denotes the constant part itself and m the mean error of the observations not yet freed of the constant part, or simply by $\frac{m}{\sqrt{\sigma}}$ if m denotes the mean error of the observations freed of the constant part (see art. 8).

16.

In artt. 12—15 we supposed that the errors x, x', x'' etc. belonged to the same kind of observations, so that their individual probabilities were expressed by the same function. But it is evident of itself that the general investigation of artt. 12—14 extends just as easily to the more general case in which the probabilities of the errors x, x', x'' etc. are expressed by different functions $\varphi x, \varphi' x', \varphi'' x''$ etc., i.e. in which those errors belong to observations of different precision or uncertainty. Let us suppose that x is the error of an observation whose mean error to be feared is $= m$; and that x', x'' etc. are the errors of other observations whose mean errors to be feared are respectively m', m'' etc. Then the mean value of the sum $xx + x'x' + x''x'' +$ etc. will be $mm + m'm' + m''m'' +$ etc.. Now if it is known from elsewhere that the quantities m, m', m'' etc. are in a given ratio, say respectively proportional to the numbers $1, \mu', \mu''$ etc., the mean value of the expression

$$\frac{xx + x'x' + x''x'' + \text{etc.}}{1 + \mu'\mu' + \mu''\mu'' + \text{etc.}}$$

will be $= mm$. But if we put a definite value of the same expression, according as chance supplies the errors x, x', x'' etc., equal to mm, the mean error to which this determination remains subject is found in a similar way to that in the preceding article

$$= \frac{\sqrt{(n^4 + n'^4 + n''^4 + \text{etc.} - m^4 - m'^4 - m''^4 - \text{etc.})}}{1 + \mu'\mu' + \mu''\mu'' + \text{etc.}}$$

where n', n'' etc. are supposed to signify, with regard to the observations to which the errors x', x'' etc. belong, the same thing as n with regard to the first observation. If, therefore, it is permissible to suppose the numbers n, n', n'' etc. proportional to m, m', m'' etc., that mean error to be feared becomes

$$= \frac{\sqrt{(n^4 - m^4)}.\sqrt{(1 + \mu'^4 + \mu''^4 + \text{etc.})}}{1 + \mu'\mu' + \mu''\mu'' + \text{etc.}}$$

But this way of determining an approximate value of m is not the most suitable. To show this more clearly, let us consider the more general expression

$$y = \frac{xx + \alpha'x'x' + \alpha''x''x'' + \text{etc.}}{1 + \alpha'\mu'\mu' + \alpha''\mu''\mu'' + \text{etc.}}$$

whose mean value will likewise be $= mm$, however the coefficients α', α'' etc. are chosen. The mean error to be feared when we suppose a definite value of y, according as chance supplies the errors x, x', x'' etc., equal to mm is found by the principles given above

$$= \frac{\sqrt{(n^4 - m^4 + \alpha'\alpha'(n'^4 - m'^4) + \alpha''\alpha''(n''^4 - m''^4) + \text{etc.})}}{1 + \alpha'\mu'\mu' + \alpha''\mu''\mu'' + \text{etc.}}$$

To make this mean error as small as possible, we shall have to put

$$\alpha' = \frac{n^4 - m^4}{n'^4 - m'^4}.\mu'\mu'$$

$$\alpha'' = \frac{n^4 - m^4}{n''^4 - m''^4}.\mu''\mu'' \text{ etc.}$$

Evidently these values cannot be worked out unless the ratio of the quantities n, n', n'' etc. to m, m', m'' etc. is also known from elsewhere: in the absence of this exact knowledge, it at least seems safest*) to suppose the former proportional to the latter (see art. 11), whence the values result

$$\alpha' = \frac{1}{\mu'\mu'}, \quad \alpha'' = \frac{1}{\mu''\mu''} \text{ etc.}$$

i.e. the coefficients α', α'' etc. must be put equal to the relative weights of the observations to which the errors x', x'' etc. belong, taking the weight of the observation to which the error x belongs as the unit. In this way, with σ denoting as above the number of the given errors, we shall have the mean value of the expression

$$\frac{xx + \alpha' x'x' + \alpha'' x''x'' + \text{etc.}}{\sigma}$$

$= mm$, and the mean error to be feared when we adopt a value of this expression determined by chance as the true value of mm

$$\frac{\sqrt{(n^4 + \alpha'\alpha' n'^4 + \alpha''\alpha'' n''^4 + \text{etc.} - \sigma m^4)}}{\sigma}$$

and hence, if indeed it is permissible to suppose n, n', n'' etc. proportional to m, m', m'',

$$\cdot = \sqrt{\frac{n^4 - m^4}{\sigma}}$$

which formula is identical with the one we found above for the case of observations of the same kind.

*) Namely, we conceive that knowledge of the quantities μ', μ'' etc. is within our power only in the case where, by the nature of the matter, the errors x, x', x'' etc. proportional to $1, \mu', \mu''$ etc. must be regarded as equally probable, or rather where

$$\varphi x = \mu'\varphi'(\mu' x) = \mu''\varphi''(\mu'' x) \text{ etc.}$$

17.

If the value of a quantity that depends on another unknown quantity has been determined by an observation that does not have absolute precision, the value of the unknown calculated from it will also be subject to error, but nothing in this determination is left to arbitrary choice. But if *several* quantities depending on the same unknown have become known through observations that are not absolutely exact, we can obtain the value of the unknown either from any one of these observations or from some combination of several observations, which can be done in infinitely many different ways. Although the value of the unknown arising in this way always remains subject to error, a greater error will be feared in one combination, a smaller in another. The case will be similar if several quantities depending simultaneously on several unknowns have been observed: according as the number of observations is equal to, less than or greater than the number of unknowns, the problem will be determined, undetermined or more than determined (at least generally speaking), and in the third case the observations can be combined in infinitely many different ways to determine the unknowns. To select from such a variety of combinations those most suited to the purpose, i.e. those giving values of the unknowns subject to the smallest errors, is certainly by far the most important problem in the application of mathematics to natural philosophy.

In Theoria motus corporum coelestium we showed how the *most probable* values of the unknowns are to be obtained if the law of probability of the errors of observations is known; and since this law by its nature remains hypothetical in almost all cases, we applied that theory to the most plausible law, where the probability of the error x is supposed proportional to the exponential quantity e^{-hhxx}. From this arose the method long used by us in calculations, especially astronomical ones, and now indeed used by most people doing calculations under the name of the method of least squares.

Later the illustrious LAPLACE, approaching the matter in another way, showed that the same principle must still be preferred to all others, whatever the law of probability of the errors, provided only that the number of observations is very large. But for a moderate number of observations the matter remained untouched, so that, if our hypothetical law is rejected, the method of least squares must be considered preferable to others only on the ground that it is particularly suited to convenient calculation.

We therefore hope that geometers will welcome our showing, in this new treatment of the subject, that the method of least squares gives the best combination of all. Not approximately, but absolutely, whatever the law of probability of the errors, whatever the number of observations, provided only that we establish the notion of mean error according to our treatment in artt. 5 and 6, rather than according to the view of the illustrious LAPLACE.

Moreover, it is appropriate to warn here in express words that all the following investigations concern only irregular errors free of a constant part, since the complete art of observation properly requires removing all causes of constant errors with the greatest care. What aids the calculus of probabilities itself can offer to someone undertaking the treatment of observations that there is good reason to suspect are not free of constant errors, we reserve for a separate investigation to be published on another occasion.

18.

PROBLEM. *Let U denote a given function of the unknown quantities V, V', V'' etc.; the mean error M to be feared in determining the value of U is required, if for V, V', V'' etc. we adopt their observed values rather than their true values, the observations being mutually independent and subject respectively to the mean errors m, m', m'' etc.*

Solution. Denoting the errors in the observed values of V, V', V'' etc. by e, e', e'' etc., the error thereby arising in the value of U can be expressed by the linear function

$$\lambda e + \lambda' e' + \lambda'' e'' + \text{etc.} = E$$

where $\lambda, \lambda', \lambda''$ etc. are the values of the differential quotients $\frac{\mathrm{d}U}{\mathrm{d}V}, \frac{\mathrm{d}U}{\mathrm{d}V'}, \frac{\mathrm{d}U}{\mathrm{d}V''}$ etc. for the true values of V, V', V'' etc., provided the observations are sufficiently exact to allow the squares and products of the errors to be neglected. From this it follows first, since the errors of the observations are supposed free of constant parts, that the mean value of E is $= 0$. Further, the mean error to be feared in the value of U will be the square root of the mean value of EE, or MM will be the mean value of the sum

$$\begin{aligned} &\lambda\lambda ee + \lambda'\lambda' e'e' + \lambda''\lambda'' e''e'' + \text{etc.} \\ &+ 2\lambda\lambda' ee' + 2\lambda\lambda'' ee'' + 2\lambda'\lambda'' e'e'' + \text{etc.} \end{aligned}$$

But the mean value of $\lambda\lambda ee$ becomes $\lambda\lambda mm$, the mean value of $\lambda'\lambda' e'e'$ becomes $= \lambda'\lambda' m'm'$ etc.; finally, the mean values of the products $2\lambda\lambda' ee'$ etc. all become $= 0$. Hence we conclude

$$M = \sqrt{(\lambda\lambda mm + \lambda'\lambda' m'm' + \lambda''\lambda'' m''m'' + \text{etc.})}$$

It will be appropriate to add some remarks to this solution.

I. Insofar as, by regarding the errors of observations as quantities of the first order, quantities of higher orders are neglected, we may also adopt in our formula for $\lambda, \lambda', \lambda''$ etc. those values of the quotients $\frac{\mathrm{d}U}{\mathrm{d}V}$ etc. that arise from the observed values of the quantities V, V', V'' etc. Whenever U is a linear function, there will evidently be no difference at all.

II. If we prefer to introduce the weights of the observations instead of their mean errors, let these, according to an arbitrary unit, be respectively p, p', p'' etc., and let P be the weight of the determination of the value of U arising from the observed values of the quantities V, V', V'' etc. We shall thus have

$$P = \frac{1}{\dfrac{\lambda\lambda}{p} + \dfrac{\lambda'\lambda'}{p'} + \dfrac{\lambda''\lambda''}{p''} + \text{etc.}}$$

III. If T is another given function of the quantities V, V', V'' etc., and if, for their true values,

$$\frac{\mathrm{d}T}{\mathrm{d}V} = \kappa, \quad \frac{\mathrm{d}T}{\mathrm{d}V'} = \kappa', \quad \frac{\mathrm{d}T}{\mathrm{d}V''} = \kappa'' \text{ etc.}$$

the error in the determination of the value of T obtained from the observed values of V, V', V'' etc. will be $= \kappa e + \kappa' e' + \kappa'' e'' + \text{etc.}$, $= E'$, and the mean error to be feared in this determination $= \sqrt{(\kappa\kappa mm + \kappa'\kappa' m'm' + \kappa''\kappa'' m''m'' + \text{etc.})}$. But the errors E, E' will evidently no longer be independent of one another. The mean value at issue is that of the product EE'. Unlike the mean value of the product ee', it will not be $= 0$, but $= \kappa\lambda mm + \kappa'\lambda' m'm' + \kappa''\lambda'' m''m'' + \text{etc.}$

IV. Our problem can also be extended to the case where the values of the quantities V, V', V'' etc. are not found directly by observations but are derived in any way from combinations of observations, provided only that their individual determinations are mutually independent, i.e. based on different observations: whenever this condition does not hold, the formula for M would become erroneous. For example, if one or another observation that served to determine the value of V had also been used to determine the value of V', the errors e and e' would no longer be mutually independent, nor therefore would the mean value of the product ee' be $= 0$. But if in such a case the connection of the quantities V, V' with the individual observations from which they have been derived is duly considered, the mean value of the product ee' can be assigned with the aid of remark III., and the formula for M can thus be made complete.

19.

Let V, V', V'' etc. be functions of the unknowns x, y, z etc., the number of the former $= \pi$, the number of the unknowns $= \rho$; and let us suppose that the values of the functions have been found by observations, either directly or indirectly, as $V = L, V' = L', V'' = L''$ etc., yet in such a way that these determinations were mutually independent. If ρ is greater than π, finding the unknowns evidently becomes an undetermined problem; the next case is where ρ is equal to π. Each of x, y, z etc. can be reduced, or conceived as reduced, to the form of functions of V, V', V'' etc., so that the values of the former can be found from the observed values of the latter, and at the same time, with the aid of the preceding article, the relative precision to be assigned to each of these determinations can be specified; finally, if ρ is less than π, each of x, y, z etc. can be reduced in infinitely many different ways to the form of functions of V, V', V'' etc., and thus their values can be obtained in infinitely many different ways. These determinations should indeed agree exactly if the observations had absolute precision; since the case is otherwise,

different methods will give different values, and likewise the determinations obtained from different combinations will have unequal precision.

Moreover, if in the second or third case the functions V, V', V'' etc. were such that $\pi - \rho + 1$ of them, or more, could be regarded as functions of the remaining ones, the problem would still be more than determined with regard to the latter functions, but undetermined with regard to the unknowns x, y, z etc.; their values could not be determined even if absolutely exact values of the functions V, V', V'' etc. were given: but we shall exclude this case from our investigation.

Whenever V, V', V'' etc. are not themselves *linear* functions of their indeterminates, this will be achieved if, in place of the original unknowns, we introduce their differences from approximate values. These may be supposed known from elsewhere. We shall denote the mean errors to be feared in the determinations $V = L, V' = L', V'' = L''$ etc. respectively by m, m', m'' etc., and the weights of the determinations by p, p', p'' etc., so that $pmm = p'm'm' = p''m''m''$ etc. We shall suppose the ratio of the mean errors to one another known, so that the weights, one of which can be chosen at will, are known. Finally, we shall put

$$(V - L)\sqrt{p} = v, \quad (V' - L')\sqrt{p'} = v',$$
$$(V'' - L'')\sqrt{p''} = v'' \text{ etc.}$$

Evidently, therefore, the case will be the same as if direct observations of equal precision, namely whose mean error is $= m\sqrt{p} = m'\sqrt{p'} = m''\sqrt{p''}$ etc., or to which the weight $= 1$ is assigned, had given

$$v = 0, \quad v' = 0, \quad v'' = 0 \text{ etc.}$$

20.

PROBLEM. *Let v, v', v'' etc. denote the following linear functions of the indeterminates x, y, z etc.*

$$\left.\begin{aligned} v &= ax + by + cz + \text{etc.} + l \\ v' &= a'x + b'y + c'z + \text{etc.} + l' \\ v'' &= a''x + b''y + c''z + \text{etc.} + l'' \text{ etc.} \end{aligned}\right\} \quad \text{(I)}$$

From all systems of coefficients $\kappa, \kappa', \kappa''$ etc. that give identically

$$\kappa v + \kappa' v' + \kappa'' v'' + \text{etc.} = x - k$$

with k a determinate quantity, i.e. independent of x, y, z etc., find the one for which $\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + $ etc. attains its smallest value.

Solution. Let us put

$$\left.\begin{aligned} av + a'v' + a''v'' + \text{etc.} &= \xi \\ bv + b'v' + b''v'' + \text{etc.} &= \eta \\ cv + c'v' + c''v'' + \text{etc.} &= \zeta \end{aligned}\right\} \quad \text{(II)}$$

etc.: then ξ, η, ζ etc. will also be linear functions of x, y, z etc., namely

$$\left.\begin{aligned} \xi &= x\Sigma aa + y\Sigma ab + z\Sigma ac + \text{etc.} + \Sigma al \\ \eta &= x\Sigma ab + y\Sigma bb + z\Sigma bc + \text{etc.} + \Sigma bl \\ \zeta &= x\Sigma ac + y\Sigma bc + z\Sigma cc + \text{etc.} + \Sigma cl \text{ etc.} \end{aligned}\right\} \quad \text{(III)}$$

(where Σaa denotes the sum $aa + a'a' + a''a'' + $ etc., and similarly for the others), and the number of ξ, η, ζ etc. is equal to the number of the indeterminates x, y, z etc., namely $= \rho$. By elimination, therefore, an equation of the following kind can be obtained[*)]

$$x = A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.}$$

in which, on substituting for ξ, η, ζ etc. their values from III, an identical equation must result. Thus, putting

$$\left.\begin{aligned} a[\alpha\alpha] + b[\alpha\beta] + c[\alpha\gamma] + \text{etc.} &= \alpha \\ a'[\alpha\alpha] + b'[\alpha\beta] + c'[\alpha\gamma] + \text{etc.} &= \alpha' \\ a''[\alpha\alpha] + b''[\alpha\beta] + c''[\alpha\gamma] + \text{etc.} &= \alpha'' \text{ etc.} \end{aligned}\right\} \quad \text{(IV)}$$

we shall necessarily have, identically,

$$\alpha v + \alpha' v' + \alpha'' v'' + \text{etc.} = x - A \quad \text{(V)}$$

[*)]The reason why we chose precisely these characters to denote the coefficients arising from such an elimination will become clear below.

This equation shows that the systems of values of the coefficients $\kappa, \kappa', \kappa''$ etc. must certainly also include $\kappa = \alpha, \kappa' = \alpha', \kappa'' = \alpha''$ etc., and that, for any system, we must have identically

$$(\kappa - \alpha)v + (\kappa' - \alpha')v' + (\kappa'' - \alpha'')v'' + \text{etc.} = A - k$$

which equation implies the following

$$\begin{aligned}
(\kappa - \alpha)a + (\kappa' - \alpha')a' + (\kappa'' - \alpha'')a'' + \text{etc.} &= 0 \\
(\kappa - \alpha)b + (\kappa' - \alpha')b' + (\kappa'' - \alpha'')b'' + \text{etc.} &= 0 \\
(\kappa - \alpha)c + (\kappa' - \alpha')c' + (\kappa'' - \alpha'')c'' + \text{etc.} &= 0 \text{ etc.}
\end{aligned}$$

Multiplying these equations respectively by $[\alpha\alpha], [\alpha\beta], [\alpha\gamma]$ etc., and adding, we obtain by (IV)

$$(\kappa - \alpha)\alpha + (\kappa' - \alpha')\alpha' + (\kappa'' - \alpha'')\alpha'' + \text{etc.} = 0$$

or, what is the same,

$$\begin{aligned}
&\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + \text{etc.} \\
&= \alpha\alpha + \alpha'\alpha' + \alpha''\alpha'' + \text{etc.} \\
&\quad + (\kappa - \alpha)^2 + (\kappa' - \alpha')^2 + (\kappa'' - \alpha'')^2 + \text{etc.}
\end{aligned}$$

whence it is evident that the sum $\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + \text{etc.}$ attains its smallest value if we put $\kappa = \alpha, \kappa' = \alpha', \kappa'' = \alpha''$ etc. Q. E. I.

Moreover, this smallest value itself is found in the following way. Equation (V) shows that

$$\begin{aligned}
\alpha a + \alpha' a' + \alpha'' a'' + \text{etc.} &= 1 \\
\alpha b + \alpha' b' + \alpha'' b'' + \text{etc.} &= 0 \\
\alpha c + \alpha' c' + \alpha'' c'' + \text{etc.} &= 0 \text{ etc.}
\end{aligned}$$

Multiplying these equations respectively by $[\alpha\alpha], [\alpha\beta], [\alpha\gamma]$ etc. and adding, we immediately obtain with the aid of equations (IV)

$$\alpha\alpha + \alpha'\alpha' + \alpha''\alpha'' + \text{etc.} = [\alpha\alpha]$$

21.

Since the observations have supplied the equations (approximately true) $v = 0, v' = 0, v'' = 0$ etc., to derive the value of the unknown x from them we must use a combination of those equations of the form

$$\kappa v + \kappa' v' + \kappa'' v'' + \text{etc.} = 0$$

which gives x the coefficient 1 and eliminates the remaining unknowns y, z etc.; by art. 18 this determination must be assigned the weight

$$= \frac{1}{\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + \text{etc.}}$$

It therefore follows from the preceding article that the most suitable determination will be the one in which we put $\kappa = \alpha, \kappa' = \alpha', \kappa'' = \alpha''$ etc. In this way x receives the value A, and evidently the same value can also (without knowing the multipliers $\alpha, \alpha', \alpha''$ etc.) be derived directly by elimination from the equations $\xi = 0, \eta = 0, \zeta = 0$ etc. The weight to be assigned to this determination will be $= \frac{1}{[\alpha\alpha]}$, or the mean error to be feared in it

$$= m\sqrt{p}[\alpha\alpha] = m'\sqrt{p'}[\alpha\alpha] = m''\sqrt{p''}[\alpha\alpha] \text{ etc.}$$

In precisely the same way, the most suitable determination of the remaining unknowns y, z etc. will give them the same values as result from elimination from the same equations $\xi = 0, \eta = 0, \zeta = 0$ etc.

Denoting the indefinite sum $vv + v'v' + v''v''$ etc., or, what amounts to the same thing, this sum

$$p(V - L)^2 + p'(V' - L')^2 + p''(V'' - L'')^2 + \text{etc.}$$

by Ω, we see that $2\xi, 2\eta, 2\zeta$ etc. are the partial differential quotients of the function Ω, namely

$$2\xi = \frac{\mathrm{d}\Omega}{\mathrm{d}x}, \quad 2\eta = \frac{\mathrm{d}\Omega}{\mathrm{d}y}, \quad 2\zeta = \frac{\mathrm{d}\Omega}{\mathrm{d}z} \text{ etc.}$$

The values of the unknowns resulting from the most suitable combination of observations, which we may conveniently call the *most plausible values*, will therefore be identical with those for which Ω attains its minimum value. Now $V - L$ expresses indefinitely the difference between the computed and observed value. Thus the most plausible values of the unknowns will be those which make the sum of the squares of the differences between the observed and computed values of the quantities V, V', V'' etc., multiplied by the weights of the observations, a minimum: a principle that we had established by a very different route in the *Theoria Motus Corporum Coelestium*. And if the relative

precision of the individual determinations is also to be assigned, we must, by indefinite elimination from equations (III), express x, y, z etc. themselves in the following form:

$$\left.\begin{aligned} x &= A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.} \\ y &= B + [\beta\alpha]\xi + [\beta\beta]\eta + [\beta\gamma]\zeta + \text{etc.} \\ z &= C + [\gamma\alpha]\xi + [\gamma\beta]\eta + [\gamma\gamma]\zeta + \text{etc.} \end{aligned}\right\} \quad \text{(VII)}$$

$$\text{etc.}$$

In this way the most plausible values of the unknowns x, y, z etc. will be respectively A, B, C etc., and the weights to be assigned to these determinations $\frac{1}{[\alpha\alpha]}, \frac{1}{[\beta\beta]}, \frac{1}{[\gamma\gamma]}$ etc., or the mean errors to be feared in them

$$\begin{aligned} &\text{for } x \ldots\ldots m\sqrt{p}[\alpha\alpha] = m'\sqrt{p'}[\alpha\alpha] = m''\sqrt{p''}[\alpha\alpha] \text{ etc.} \\ &\text{for } y \ldots\ldots m\sqrt{p}[\beta\beta] = m'\sqrt{p'}[\beta\beta] = m''\sqrt{p''}[\beta\beta] \text{ etc.} \\ &\text{for } z \ldots\ldots m\sqrt{p}[\gamma\gamma] = m'\sqrt{p'}[\gamma\gamma] = m''\sqrt{p''}[\gamma\gamma] \text{ etc.} \end{aligned}$$

$$\text{etc.}$$

in agreement with what we taught in the *Theoria Motus Corporum Coelestium*.

22.

It will be useful to discuss separately, in a few words, the simplest and at the same time most frequent case, in which there is only one unknown and $V = x, V' = x, V'' = x$ etc. We shall then have $a = \sqrt{p}, a' = \sqrt{p'}, a'' = \sqrt{p''}$ etc., $l = -L\sqrt{p}, l' = -L'\sqrt{p'}, l'' = -L''\sqrt{p''}$ etc., and hence

$$\xi = (p + p' + p'' + \text{etc.})x - (pL + p'L' + p''L'' + \text{etc.})$$

From this, further,

$$[\alpha\alpha] = \frac{1}{p + p' + p'' + \text{etc.}}$$

$$A = \frac{pL + p'L' + p''L'' + \text{etc.}}{p + p' + p'' + \text{etc.}}$$

If, therefore, several observations of unequal precision, whose weights are respectively p, p', p'' etc., have given the value of the same quantity from the first $= L$, from the second $= L'$, from the third $= L''$ etc., its most plausible value will be

$$= \frac{pL + p'L' + p''L'' + \text{etc.}}{p + p' + p'' + \text{etc.}}$$

and the weight of this determination $= p+p'+p''$ etc. If all observations have equal precision, the most plausible value will be

$$= \frac{L + L' + L'' + \text{etc.}}{\pi}$$

that is, equal to the arithmetic mean of the observed values, and the weight of this determination $= \pi$, taking the weight of the observations as the unit.

THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS

Second Part

BY

CARL FRIEDRICH GAUSS

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THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS.

Second Part.

23.

Several investigations still remain by which the preceding theory will be both illustrated and extended.

First of all, we must investigate whether the elimination by means of which the indeterminates x, y, z etc. are to be expressed through ξ, η, ζ etc. is always possible. Since the number of the former equals that of the latter, the theory of elimination in linear equations gives the following result about this elimination. If ξ, η, ζ etc. are mutually independent, it will certainly be possible; otherwise, impossible. Let us suppose for the moment that ξ, η, ζ etc. are not mutually independent. Suppose instead that an identity exists between them

$$0 = F\xi + G\eta + H\zeta + \text{etc.} + K$$

We shall therefore have

$$\begin{aligned} F\Sigma aa + G\Sigma ab + H\Sigma ac + \text{etc.} &= 0 \\ F\Sigma ab + G\Sigma bb + H\Sigma bc + \text{etc.} &= 0 \\ F\Sigma ac + G\Sigma bc + H\Sigma cc + \text{etc.} &= 0 \end{aligned}$$

etc., and also

$$F\Sigma al + G\Sigma bl + H\Sigma cl + \text{etc.} = -K$$

Putting, further,

$$\left.\begin{aligned} aF + bG + cH + \text{etc.} &= \theta \\ a'F + b'G + c'H + \text{etc.} &= \theta' \\ a''F + b''G + c''H + \text{etc.} &= \theta'' \end{aligned}\right\} \quad \text{(I)}$$

etc., we obtain

$$\begin{aligned} a\theta + a'\theta' + a''\theta'' + \text{etc.} &= 0 \\ b\theta + b'\theta' + b''\theta'' + \text{etc.} &= 0 \\ c\theta + c'\theta' + c''\theta'' + \text{etc.} &= 0 \end{aligned}$$

etc., and also

$$l\theta + l'\theta' + l''\theta'' + \text{etc.} = -K$$

Multiplying equations (I) respectively by $\theta, \theta', \theta''$ etc. and adding, we therefore obtain:

$$0 = \theta\theta + \theta'\theta' + \theta''\theta'' + \text{etc.}$$

This equation evidently cannot hold unless $\theta = 0, \theta' = 0, \theta'' = 0$ etc. hold simultaneously. From this we first conclude that necessarily $K = 0$. Equations (I) then show that the functions v, v', v'' etc. are such that their values do not change if the values of the quantities x, y, z etc. receive increments or decrements respectively proportional to F, G, H etc.; evidently the same will hold for the functions V, V', V'' etc. Our supposition can therefore hold only in a case in which even from the exact values of the quantities V, V', V'' etc. it would have been impossible to determine the values of the unknowns x, y, z etc., that is, in which the problem would by its nature have been indeterminate, a case we excluded from our investigation.

24.

Let β, β', β'' etc. denote multipliers which have the same relation to the indeterminate y as $\alpha, \alpha', \alpha''$ etc. have to x; namely, let

$$\begin{aligned} a[\beta\alpha] + b[\beta\beta] + c[\beta\gamma] + \text{etc.} &= \beta \\ a'[\beta\alpha] + b'[\beta\beta] + c'[\beta\gamma] + \text{etc.} &= \beta' \\ a''[\beta\alpha] + b''[\beta\beta] + c''[\beta\gamma] + \text{etc.} &= \beta'' \end{aligned}$$

etc., so that identically

$$\beta v + \beta' v' + \beta'' v'' + \text{etc.} = y - B$$

Similarly, let $\gamma, \gamma', \gamma''$ etc. be analogous multipliers with respect to the indeterminate z, namely

$$\begin{aligned} a[\gamma\alpha] + b[\gamma\beta] + c[\gamma\gamma] + \text{etc.} &= \gamma \\ a'[\gamma\alpha] + b'[\gamma\beta] + c'[\gamma\gamma] + \text{etc.} &= \gamma' \\ a''[\gamma\alpha] + b''[\gamma\beta] + c''[\gamma\gamma] + \text{etc.} &= \gamma'' \end{aligned}$$

etc., so that identically

$$\gamma v + \gamma' v' + \gamma'' v'' + \text{etc.} = z - C$$

and so on. In this way, just as we already found in art. 20 that

$$\Sigma\alpha a = 1, \quad \Sigma\alpha b = 0, \quad \Sigma\alpha c = 0, \quad \text{etc., and also} \quad \Sigma\alpha l = -A$$

we shall also have

$$\begin{aligned} &\Sigma\beta a = 0, \quad \Sigma\beta b = 1, \quad \Sigma\beta c = 0, \quad \text{etc., and} \quad \Sigma\beta l = -B \\ &\Sigma\gamma a = 0, \quad \Sigma\gamma b = 0, \quad \Sigma\gamma c = 1, \quad \text{etc., and} \quad \Sigma\gamma l = -C \end{aligned}$$

and so on. Likewise, just as art. 20 gave $\Sigma\alpha\alpha = [\alpha\alpha]$, we shall also have

$$\Sigma\beta\beta = [\beta\beta], \quad \Sigma\gamma\gamma = [\gamma\gamma] \quad \text{etc.}$$

Multiplying, further, the values of $\alpha, \alpha', \alpha''$ etc. (art. 20. IV) respectively by β, β', β'' etc. and adding, we obtain

$$\alpha\beta + \alpha'\beta' + \alpha''\beta'' \quad \text{etc.} = [\alpha\beta], \quad \text{or} \quad \Sigma\alpha\beta = [\alpha\beta]$$

But multiplying the values of β, β', β'' etc. respectively by $\alpha, \alpha', \alpha''$ etc. and adding gives, in the same way,

$$\alpha\beta + \alpha'\beta' + \alpha''\beta'' + \text{etc.} = [\beta\alpha], \quad \text{hence} \quad [\alpha\beta] = [\beta\alpha]$$

In precisely the same way we obtain

$$[\alpha\gamma] = [\gamma\alpha] = \Sigma\alpha\gamma, \quad [\beta\gamma] = [\gamma\beta] = \Sigma\beta\gamma \quad \text{etc.}$$

25.

Let $\lambda, \lambda', \lambda''$ etc. further denote the values of the functions v, v', v'' etc. obtained when for x, y, z etc. we substitute their most plausible values A, B, C etc., namely

$$aA + bB + cC + \text{etc.} + l = \lambda$$
$$a'A + b'B + c'C + \text{etc.} + l' = \lambda'$$
$$a''A + b''B + c''C + \text{etc.} + l'' = \lambda''$$

etc.; let us put, in addition,

$$\lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.} = M$$

so that M is the value of the function Ω corresponding to the most plausible values of the indeterminates, and hence, by what we proved in art. 20, the minimum value of this function. Then $a\lambda + a'\lambda' + a''\lambda'' + \text{etc.}$ will be the value of ξ corresponding to the values $x = A, y = B, z = C$ etc., and therefore $= 0$, that is, we shall have

$$\Sigma a\lambda = 0$$

and similarly

$$\Sigma b\lambda = 0, \quad \Sigma c\lambda = 0 \quad \text{etc.};$$
$$\text{and also} \quad \Sigma\alpha\lambda = 0, \quad \Sigma\beta\lambda = 0, \quad \Sigma\gamma\lambda = 0 \quad \text{etc.}$$

Finally, multiplying the expressions for $\lambda, \lambda', \lambda''$ etc. respectively by $\lambda, \lambda', \lambda''$ etc. and adding, we obtain $l\lambda + l'\lambda' + l''\lambda'' + \text{etc.} = \lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.}$, or

$$\Sigma l\lambda = M$$

26.

In the equation $v = ax + by + cz + \text{etc.} + l$, substituting expressions VII of art. 21 for x, y, z etc. and applying the reductions evident from what precedes gives

$$v = \alpha\xi + \beta\eta + \gamma\zeta + \text{etc.} + \lambda$$

and in the same way we shall have identically

$$v' = \alpha'\xi + \beta'\eta + \gamma'\zeta + \text{etc.} + \lambda'$$
$$v'' = \alpha''\xi + \beta''\eta + \gamma''\zeta + \text{etc.} + \lambda''$$

etc. Multiplying either these equations or equations I of art. 20 respectively by $\lambda, \lambda', \lambda''$ etc. and adding shows that identically

$$\lambda v + \lambda' v' + \lambda'' v'' + \text{etc.} = M$$

27.

The function Ω can be expressed in indefinitely many forms, which it will be worth developing. First, by squaring equations I of art. 20 and adding, we obtain immediately

$$\begin{aligned}\Omega = {} & xx\Sigma aa + yy\Sigma bb + zz\Sigma cc + \text{etc.} \\ & + 2xy\Sigma ab + 2xz\Sigma ac + 2yz\Sigma bc + \text{etc.} \\ & + 2x\Sigma al + 2y\Sigma bl + 2z\Sigma cl + \text{etc.} + \Sigma ll\end{aligned}$$

which is the *first* form.

Multiplying the same equations respectively by v, v', v'' etc. and adding, we obtain:

$$\Omega = \xi x + \eta y + \zeta z + \text{etc.} + lv + l'v' + l''v'' + \text{etc.}$$

and hence, substituting for v, v', v'' etc. the expressions given in the preceding article,

$$\Omega = \xi x + \eta y + \zeta z + \text{etc.} - A\xi - B\eta - C\zeta - \text{etc.} + M$$

or

$$\Omega = \xi(x - A) + \eta(y - B) + \zeta(z - C) + \text{etc.} + M$$

which is the *second* form.

Substituting expressions VII of art. 21 for $x - A, y - B, z - C$ etc. in the second form, we obtain the *third* form:

$$\begin{aligned}\Omega = {} & [\alpha\alpha]\xi\xi + [\beta\beta]\eta\eta + [\gamma\gamma]\zeta\zeta + \text{etc.} \\ & + 2[\alpha\beta]\xi\eta + 2[\alpha\gamma]\xi\zeta + 2[\beta\gamma]\eta\zeta + \text{etc.} + M\end{aligned}$$

To these we may add the *fourth* form, which follows of itself from the third form and the formulas of the preceding article:

$$\Omega = (v - \lambda)^2 + (v' - \lambda')^2 + (v'' - \lambda'')^2 + \text{etc.} + M, \quad \text{or}$$

$$\Omega = M + \Sigma(v - \lambda)^2$$

This form puts the condition for a minimum directly before our eyes.

28.

Let e, e', e'' etc. be the errors committed in the observations that gave $V = L, V' = L' \quad V'' = L''$ etc.; that is, let the true values of the functions V, V', V'' etc. be respectively $L - e, L' - e', L'' - e''$ etc., and hence the true values of v, v', v'' etc. respectively $-e\sqrt{p}, -e'\sqrt{p'}, -e''\sqrt{p''}$ etc. The true value of x will therefore be

$$= A - \alpha e\sqrt{p} - \alpha' e'\sqrt{p'} - \alpha'' e''\sqrt{p''} - \text{etc.}$$

or, for the error committed in the best determination of the value of x, which it is convenient to denote by Ex, we have

$$= \alpha e\sqrt{p} + \alpha' e'\sqrt{p'} + \alpha'' e''\sqrt{p''} + \text{etc.}$$

Similarly, the error committed in the best determination of the value of y, which we shall denote by Ey, will be

$$= \beta e\sqrt{p} + \beta' e'\sqrt{p'} + \beta'' e''\sqrt{p''} + \text{etc.}$$

The mean value of the square $(Ex)^2$ is found to be

$$= mmp(\alpha\alpha + \alpha'\alpha' + \alpha''\alpha'' + \text{etc.}) = mmp[\alpha\alpha]$$

and for the mean value of the square $(Ey)^2$ we similarly find $= mmp[\beta\beta]$ etc., as we have already shown above. We may now also assign the mean value of the *product* $Ex.Ey$, which is found to be

$$= mmp(\alpha\beta + \alpha'\beta' + \alpha''\beta'' + \text{etc.}) = mmp[\alpha\beta]$$

These statements can also be expressed neatly as follows. The mean values of the squares $(Ex)^2, (Ey)^2$ etc. are respectively equal to the products of $\frac{1}{2}mmp$ and the second partial derivatives

$$\frac{dd\Omega}{d\xi^2}, \quad \frac{dd\Omega}{d\eta^2} \quad \text{etc.}$$

and the mean value of a product such as $Ex.Ey$ is equal to the product of $\frac{1}{2}mmp$ and the derivative $\frac{dd\Omega}{d\xi.d\eta}$: provided, of course, that Ω is considered as a function of the indeterminates ξ, η, ζ etc.

29.

Let t denote a given linear function of the quantities x, y, z etc.; say,

$$t = fx + gy + hz + \text{etc.} + k$$

The value of t obtained from the most plausible values of x, y, z etc. will therefore be $= fA + gB + hC + \text{etc.} + k$, which we shall denote by K. If this is adopted as the true value of t, an error is committed, which will be

$$= fEx + gEy + hEz + \text{etc.}$$

and will be denoted by Et. Clearly, the mean value of this error is $= 0$, or the error will be free of a constant part. But the mean value of the square $(Et)^2$, or the mean value of the sum

$$\begin{array}{llll} ff(Ex)^2 & +2fg\,Ex.Ey & +2fh\,Ex.Ez & +\text{etc.} \\ & +gg(Ey)^2 & +2gh\,Ey.Ez & +\text{etc.} \\ & & +hh(Ez)^2 & +\text{etc. etc.} \end{array}$$

is, by what we set out in the preceding article, equal to the product of mmp and the sum

$$\begin{array}{llll} ff[\alpha\alpha] & +2fg[\alpha\beta] & +2fh[\alpha\gamma] & +\text{etc.} \\ & +gg[\beta\beta] & +2gh[\beta\gamma] & +\text{etc.} \\ & & +hh[\gamma\gamma] & +\text{etc. etc.} \end{array}$$

or to the product of mmp and the value of the function $\Omega - M$ obtained by the substitutions

$$\xi = f, \quad \eta = g, \quad \zeta = h \quad \text{etc.}$$

Thus, denoting this particular value of the function $\Omega - M$ by ω, the mean error to be feared while we adhere to the determination $t = K$ will be $= m\sqrt{p\omega}$, or the weight of this determination will be $= \frac{1}{\omega}$.

Since we have identically

$$\Omega - M = (x - A)\xi + (y - B)\eta + (z - C)\zeta + \text{etc.}$$

it is clear that ω is also equal to the particular value of the expression

$$(x - A)f + (y - B)g + (z - C)h + \text{etc.}$$

or to the particular value of $t - K$ obtained when the indeterminates x, y, z etc. are assigned those values that correspond to the values ξ, η, ζ etc. being f, g, h etc.

Finally, we observe that if t is expressed identically as a function of ξ, η, ζ etc., its constant part must be $= K$. Thus, if identically

$$t = F\xi + G\eta + H\zeta + \text{etc.} + K, \quad \text{then} \quad \omega = fF + gG + hH + \text{etc.}$$

30.

The function Ω attains its *absolute minimum* value M, as we saw above, on putting $x = A, y = B, z = C$ etc., or $\xi = 0, \eta = 0, \zeta = 0$ etc. But if one of these quantities has already been assigned a *different* value, e.g. $x = A + \Delta$, then, as the others vary, Ω can attain a *relative minimum* value, which is clearly obtained by means of the equations

$$x = A + \Delta, \quad \frac{d\Omega}{dy} = 0, \quad \frac{d\Omega}{dz} = 0 \quad \text{etc.}$$

We must therefore have $\eta = 0, \zeta = 0$ etc., and hence the relation gives

$$x = A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.}, \quad \xi = \frac{\Delta}{[\alpha\alpha]}$$

At the same time we shall have

$$y = B + \frac{[\alpha\beta]\Delta}{[\alpha\alpha]}, \quad z = C + \frac{[\alpha\gamma]\Delta}{[\alpha\alpha]} \quad \text{etc.}$$

The relative minimum value of Ω is then $= [\alpha\alpha]\xi\xi + M = M + \frac{\Delta\Delta}{[\alpha\alpha]}$. Conversely, we conclude from this that if the value of Ω must not exceed the prescribed bound $M + \mu\mu$, the value of x must necessarily lie between the bounds $A - \mu\sqrt{[\alpha\alpha]}$ and $A + \mu\sqrt{[\alpha\alpha]}$. It deserves notice that $\mu\sqrt{[\alpha\alpha]}$ is equal to the mean error to be feared in the most plausible value of x if we put $\mu = m\sqrt{p}$, that is, if μ is equal to the mean error of observations to which the weight $= 1$ is assigned.

Let us investigate more generally the minimum value of Ω that can occur for a given value of t, where, as in the preceding article, t denotes the linear function $fx + gy + hz + \text{etc.} + k$ and its most plausible value is $= K$: let the prescribed value of t be denoted by $K + \kappa$. From the theory of maxima and minima, it is known that the solution of the problem is to be sought from the equations

$$\frac{d\Omega}{dx} = \theta\frac{dt}{dx}$$
$$\frac{d\Omega}{dy} = \theta\frac{dt}{dy}$$
$$\frac{d\Omega}{dz} = \theta\frac{dt}{dz} \quad \text{etc.}$$

or $\xi = \theta f, \eta = \theta g, \zeta = \theta h$ etc., where θ denotes a multiplier that is still undetermined. Thus, if, as in the preceding article, we put *identically*

$$t = F\xi + G\eta + H\zeta + \text{etc.} + K$$

we shall have

$$K + \kappa = \theta(fF + gG + hH + \text{etc.}) + K, \quad \text{or}$$
$$\theta = \frac{\kappa}{\omega}$$

taking ω in the same sense as in the preceding article. And since $\Omega - M$ is identically a homogeneous function of the second degree in the indeterminates ξ, η, ζ etc., it follows of itself that its value for $\xi = \theta f, \eta = \theta g, \zeta = \theta h$ etc. becomes $= \theta\theta\omega$, and hence that the minimum value that Ω can attain for $t = K + \kappa$ becomes $= M + \theta\theta\omega = M + \frac{\kappa\kappa}{\omega}$. Conversely, if Ω must not exceed a prescribed value $M + \mu\mu$, the value of t must necessarily lie between the bounds $K - \mu\sqrt{\omega}$ and $K + \mu\sqrt{\omega}$, where $\mu\sqrt{\omega}$ is equal to the mean error to be feared in the most plausible determination of t if for μ we take the mean error of observations to which the weight $= 1$ is assigned.

31.

Whenever the number of quantities x, y, z etc. is somewhat larger, the numerical determination of the values A, B, C etc. from the equations $\xi = 0$, $\eta = 0$. $\zeta = 0$ etc. by ordinary elimination becomes rather laborious. For this reason, in Theoria Motus Corporum Coelestium, art. 182, we indicated a special algorithm and explained it at length in the *Disquisitione de elementis ellipticis Palladis* (Comm. recent. Soc. Gotting Vol. I), by which this labour is made as simple as the matter permits. Namely, the function Ω is to be brought into a form such as:

$$\frac{u^0 u^0}{\mathfrak{A}^0} + \frac{u' u'}{\mathfrak{B}'} + \frac{u'' u''}{\mathfrak{C}''} + \frac{u''' u'''}{\mathfrak{D}'''} + \text{etc.} + M$$

Here the divisors $\mathfrak{A}^0, \mathfrak{B}', \mathfrak{C}'', \mathfrak{D}'''$ etc. are determinate quantities, while u^0, u', u'', u''' etc. are linear functions of x, y, z etc.; the second, u', however, is free of x, the third, u'', free of x and y, the fourth, u''', free of x, y and z, and so on, so that the last, $u^{(\pi-1)}$, involves only the last of the indeterminates x, y, z etc. Finally, the coefficients by which x, y, z etc. are respectively multiplied in u^0, u', u'' etc. are respectively equal to $\mathfrak{A}^0, \mathfrak{B}', \mathfrak{C}''$ etc. Once this has been done, we must put $u^0 = 0, u' = 0, u'' = 0, u''' = 0$ etc., from which the values of the unknowns x, y, z etc. will be obtained most conveniently in reverse order. It seems unnecessary to repeat here the algorithm itself by which this transformation of the function Ω is carried out.

But the indeterminate elimination, by means of which the weights of these determinations must be found, requires a much more lengthy calculation still. Indeed, the weight of the determination of the last unknown (which alone enters the last $u^{(\pi-1)}$) is easily found, from what has been proved in Theoria Motus Corporum Coelestium, to be equal to the last term in the series of divisors $\mathfrak{A}^0, \mathfrak{B}', \mathfrak{C}''$ etc. Hence several calculators, to avoid that laborious elimination when other means were lacking, resorted to repeating the algorithm of which we have spoken several times, changing the order of the quantities x, y, z etc. so that each in turn occupied the last place. We therefore hope that geometers will welcome our presenting here a new method of calculating the weights of the determinations, drawn from a deeper investigation of the subject, which seems to leave nothing further to be desired.

32.

Let us therefore put (I)

$$\begin{array}{rcccccc} u^0 & = & \mathfrak{A}^0 x & +\mathfrak{B}^0 y & +\mathfrak{C}^0 z & +\text{etc.} & +\mathfrak{L}^0 \\ u' & = & & \mathfrak{B}' y & +\mathfrak{C}' z & +\text{etc.} & +\mathfrak{L}' \\ u'' & = & & & \mathfrak{C}'' z & +\text{etc.} & +\mathfrak{L}'' \\ \text{etc.} & & & & & & \end{array}$$

Then we shall have identically

$$\begin{aligned}\tfrac{1}{2}d\Omega &= \xi\,dx + \eta\,dy + \zeta\,dz + \text{etc.}\\ &= \frac{u^0 du^0}{\mathfrak{A}^0} + \frac{u' du'}{\mathfrak{B}'} + \frac{u'' du''}{\mathfrak{C}''} + \text{etc.}\\ &= u^0\left(dx + \frac{\mathfrak{B}^0}{\mathfrak{A}^0}dy + \frac{\mathfrak{C}^0}{\mathfrak{A}^0}dz + \text{etc.}\right)\\ &\qquad + u'\left(dy + \frac{\mathfrak{C}'}{\mathfrak{B}'}dz + \text{etc.}\right) + u''(dz + \text{etc.}) + \text{etc.}\end{aligned}$$

from which we conclude (II)

$$\begin{aligned}\xi &= u^0\\ \eta &= \frac{\mathfrak{B}^0}{\mathfrak{A}^0}u^0 + u'\\ \zeta &= \frac{\mathfrak{C}^0}{\mathfrak{A}^0}u^0 + \frac{\mathfrak{C}'}{\mathfrak{B}'}u' + u''\\ &\text{etc.}\end{aligned}$$

Suppose that from this the following formulas are derived (III)

$$\begin{aligned}u^0 &= \xi\\ u' &= A'\xi + \eta\\ u'' &= A''\xi + B''\eta + \zeta\\ &\text{etc.}\end{aligned}$$

Now, from the total differential of the equation

$$\Omega = \xi(x - A) + \eta(y - B) + \zeta(z - C) + \text{etc.} + M$$

subtracting the equation

$$\tfrac{1}{2}d\Omega = \xi\,dx + \eta\,dy + \zeta\,dz + \text{etc.}$$

gives

$$\tfrac{1}{2}d\Omega = (x - A)d\xi + (y - B)d\eta + (z - C)d\zeta + \text{etc.}$$

This expression must be identical with the following one derived from III:

$$\begin{aligned}&\frac{u^0}{\mathfrak{A}^0}.d\xi + \frac{u'}{\mathfrak{B}'}(A'd\xi + d\eta)\\ &\qquad + \frac{u''}{\mathfrak{C}''}(A''d\xi + B''d\eta + d\zeta) + \text{etc.}\end{aligned}$$

From this we conclude (IV)

$$\begin{array}{rcccccc} x & = & \frac{u^0}{\mathfrak{A}^0} & +A'.\frac{u'}{\mathfrak{B}'} & +A''.\frac{u''}{\mathfrak{C}''} & +\text{etc.} + A \\ y & = & & \frac{u'}{\mathfrak{B}'} & +B''.\frac{u''}{\mathfrak{C}''} & +\text{etc.} + B \\ z & = & & & \frac{u''}{\mathfrak{C}''} & +\text{etc.} + C \\ \text{etc.} & & & & & & \end{array}$$

Substituting in these expressions for u^0, u', u'' etc. their values taken from III will complete the indeterminate elimination. In particular, to determine the weights we shall have (V)

$$\begin{array}{rcccc} [\alpha\alpha] = & \frac{1}{\mathfrak{A}^0} + \frac{A'A'}{\mathfrak{B}'} & + \frac{A''A''}{\mathfrak{C}''} & + \frac{A'''A'''}{\mathfrak{D}'''} & + \text{etc.} \\ [\beta\beta] = & \frac{1}{\mathfrak{B}'} + \frac{B''B''}{\mathfrak{C}''} & + \frac{B'''B'''}{\mathfrak{D}'''} & + \text{etc.} & \\ [\gamma\gamma] = & \frac{1}{\mathfrak{C}''} + \frac{C'''C'''}{\mathfrak{D}'''} & + \text{etc.} & & \\ \text{etc.} & & & & \end{array}$$

The simplicity of these formulas leaves nothing to be desired. Equally simple formulas also result for the remaining coefficients $[\alpha\beta], [\alpha\gamma], [\beta\gamma]$ etc., but since their use is less frequent, we refrain from giving them here.

33.

Because of the importance of the matter, and so that everything may be ready for calculation, it seemed appropriate also to write down here explicit formulas for determining the coefficients A', A'', A''' etc., B'', B''' etc. etc. This calculation can be carried out in two ways, since identities must result both when the values of u^0, u', u'' etc. taken from III are substituted in II, and when the values of ξ, η, ζ etc. from II are substituted in III. The first way supplies the following systems of formulas:

$$\frac{\mathfrak{B}^0}{\mathfrak{A}^0} + A' = 0$$

$$\frac{\mathfrak{C}^0}{\mathfrak{A}^0} + \frac{\mathfrak{C}'}{\mathfrak{B}'}.A' + A'' = 0$$

$$\frac{\mathfrak{D}^0}{\mathfrak{A}^0} + \frac{\mathfrak{D}'}{\mathfrak{B}'}.A' + \frac{\mathfrak{D}''}{\mathfrak{C}''}.A'' + A''' = 0$$

etc., from which A', A'', A''' etc. are found.

$$\frac{\mathfrak{C}'}{\mathfrak{B}'} + B'' = 0$$

$$\frac{\mathfrak{D}'}{\mathfrak{B}'} + \frac{\mathfrak{D}''}{\mathfrak{C}''}.B'' + B''' = 0$$

etc., from which B'', B''' etc. are found.

$$\frac{\mathfrak{D}''}{\mathfrak{C}''} + C''' = 0$$

etc., from which C''' etc. are found. And so on.

The other way gives these formulas:

$$\mathfrak{A}^0 A' + \mathfrak{B}^0 = 0$$

from which A' is obtained.

$$\begin{array}{llll} \mathfrak{A}^0 A'' & +\mathfrak{B}^0 B'' & +\mathfrak{C}^0 & = 0 \\ & \mathfrak{B}' B'' & +\mathfrak{C}' & = 0 \end{array}$$

from which B'' and A'' are found.

$$\begin{array}{llllll} \mathfrak{A}^0 A''' & +\mathfrak{B}^0 B''' & +\mathfrak{C}^0 C''' & +\mathfrak{D}^0 & = 0 \\ & \mathfrak{B}' B''' & +\mathfrak{C}' C''' & +\mathfrak{D}' & = 0 \\ & & \mathfrak{C}'' C''' & +\mathfrak{D}'' & = 0 \end{array}$$

from which C''', B''', A''' are found. And so on.

The two ways are almost equally convenient if the weights of the determinations of all the quantities x, y, z etc. are wanted; but whenever only one or two of the quantities $[\alpha\alpha], [\beta\beta], [\gamma\gamma]$ etc. are required, the first system will clearly be far preferable.

Moreover, combining equations I with IV leads to the same formulas and also supplies two ways of calculating the most plausible values A, B, C etc. themselves; namely, *first*,

$$A = -\frac{\mathfrak{L}^0}{\mathfrak{A}^0} - A'\frac{\mathfrak{L}'}{\mathfrak{B}'} - A''\frac{\mathfrak{L}''}{\mathfrak{C}''} - A'''\frac{\mathfrak{L}'''}{\mathfrak{D}'''} - \text{etc.}$$

$$B = -\frac{\mathfrak{L}'}{\mathfrak{B}'} - B''\frac{\mathfrak{L}''}{\mathfrak{C}''} - B'''\frac{\mathfrak{L}'''}{\mathfrak{D}'''} - \text{etc.}$$

$$C = -\frac{\mathfrak{L}''}{\mathfrak{C}''} - C'''\frac{\mathfrak{L}'''}{\mathfrak{D}'''} - \text{etc.}$$

$$\text{etc.}$$

The other calculation is identical with the ordinary one, in which we put $u^0 = 0, u' = 0, u'' = 0$ etc.

34.

What we set out in art. 32 consists only of special cases of a more general theorem, which is as follows:

THEOREM. *Let t denote the following linear function of the indeterminates x, y, z etc.*

$$t = fx + gy + hz + \text{etc.} + k$$

which, when transformed into a function of the indeterminates u^0, u', u'' etc., becomes

$$t = k^0u^0 + k'u' + k''u'' + \text{etc.} + K$$

Then K will be the most plausible value of t, and the weight of this determination will be

$$= \frac{1}{\mathfrak{A}^0k^0k^0 + \mathfrak{B}'k'k' + \mathfrak{C}''k''k'' + \text{etc.}}$$

Proof. The first part of the theorem is clear because the most plausible value of t must correspond to the values $u^0 = 0, u' = 0, u'' = 0$ etc. To prove the latter part, we observe that, since $\frac{1}{2}d\Omega = \xi\,dx + \eta\,dy + \zeta\,dz + \text{etc.}$ and $dt = f\,dx + g\,dy + h\,dz + \text{etc.}$, the following holds for $\xi = f.\quad \eta = g, \zeta = h$ etc., independently of the values of the differentials dx, dy, dz etc.:

$$d\Omega = 2dt$$

It therefore follows that, for the same values $\xi = f, \eta = g, \zeta = h$ etc.,

$$\frac{u^0}{\mathfrak{A}^0}du^0 + \frac{u'}{\mathfrak{B}'}du' + \frac{u''}{\mathfrak{C}''}.du'' + \text{etc.}$$
$$= k^0du^0 + k'du' + k''du'' + \text{etc.}$$

It is now readily seen that if dx, dy, dz etc. are independent of one another, then du^0, du', du'' etc. are also independent of one another; from this we conclude that for $\xi = f, \eta = g, \zeta = h$ etc.,

$$u^0 = \mathfrak{A}^0 k^0, \quad u' = \mathfrak{B}' k', \quad u'' = \mathfrak{C}'' k'' \quad \text{etc.}$$

The value of Ω corresponding to the same values will therefore be

$$= \mathfrak{A}^0 k^0 k^0 + \mathfrak{B}' k' k' + \mathfrak{C}'' k'' k'' + \text{etc.} + M$$

from which the truth of our theorem follows directly by art. 29.

If, moreover, we wish to carry out the transformation of the function t directly, i. e. without knowing substitutions IV of art. 32, the following formulas are available:

$$\begin{aligned} f &= \mathfrak{A}^0 k^0 \\ g &= \mathfrak{B}^0 k^0 + \mathfrak{B}' k' \\ h &= \mathfrak{C}^0 k^0 + \mathfrak{C}' k' + \mathfrak{C}'' k'' \quad \text{etc.}, \end{aligned}$$

from which the coefficients k^0, k', k'' etc. will be determined successively, and finally we shall have

$$K = k - \mathfrak{L}^0 k^0 - \mathfrak{L}' k' - \mathfrak{L}'' k'' - \text{etc.}$$

35.

The following problem deserves a separate treatment, both because of its practical usefulness and because of the elegance of its solution.

To find the changes in the most plausible values of the unknowns produced by the addition of a new equation, and the weights of the new determinations.

We shall retain the notation used above, so that the original equations, reduced to weight = 1, are these: $v = 0, v' = 0, v'' = 0$ etc.; for the indefinite sum we write $vv + v'v' + v''v''$ etc. $= \Omega$; and ξ, η, ζ etc. are the partial differential quotients

$$\frac{d\Omega}{2dx}, \quad \frac{d\Omega}{2dy}, \quad \frac{d\Omega}{2dz} \quad \text{etc.}$$

and finally indefinite elimination gives

$$\left.\begin{aligned} x &= A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.} \\ y &= B + [\alpha\beta]\xi + [\beta\beta]\eta + [\beta\gamma]\zeta + \text{etc.} \\ z &= C + [\alpha\gamma]\xi + [\beta\gamma]\eta + [\gamma\gamma]\zeta + \text{etc.} \end{aligned}\right\} \quad \text{(I)}$$

Now let us suppose that a new equation $v^* = 0$ is added (approximately true, and with weight = 1), and let us investigate how much this will change both the most plausible values of the unknowns A, B, C etc. and the coefficients $[\alpha\alpha], [\alpha\beta]$ etc.

Let us put $\Omega + v^*v^* = \Omega^*$, $\frac{d\Omega^*}{2dx} = \xi^*$, $\frac{d\Omega^*}{2dy} = \eta^*$, $\frac{d\Omega^*}{2dz} = \zeta^*$ etc., and suppose that elimination then gives

$$x = A^* + [\alpha\alpha^*]\xi^* + [\alpha\beta^*]\eta^* + [\alpha\gamma^*]\zeta^* \quad \text{etc.}$$

Finally, let

$$v^* = fx + gy + hz + \text{etc.} + k$$

so that, on substituting for x, y, z etc. their values from (I), we obtain

$$v^* = F\xi + G\eta + H\zeta + \text{etc.} + K$$

and let us put

$$Ff + Gg + Hh + \text{etc.} = \omega$$

Clearly K will be the most plausible value of the function v^* as far as it follows from the original equations, without regard to the value 0 supplied by the additional observation, and $\frac{1}{\omega}$ will be the weight of this determination.

We now have

$$\xi^* = \xi + fv^*, \quad \eta^* = \eta + gv^*, \quad \zeta^* = \zeta + hv^* \quad \text{etc.}$$

and therefore

$$\begin{aligned} &F\xi^* + G\eta^* + H\zeta^* + \text{etc.} + K \\ &\quad = v^*(1 + Ff + Gg + Hh + \text{etc.}) \end{aligned}$$

or

$$v^* = \frac{F\xi^* + G\eta^* + H\zeta^* + \text{etc.} + K}{1+\omega}$$

Similarly we obtain

$$\begin{aligned} x &= A + [\alpha\alpha]\xi^* + [\alpha\beta]\eta^* + [\alpha\gamma]\zeta^* + \text{etc.} \\ &\quad - v^*(f[\alpha\alpha] + g[\alpha\beta] + h[\alpha\gamma] + \text{etc.}) \\ &= A + [\alpha\alpha]\xi^* + [\alpha\beta]\eta^* + [\alpha\gamma]\zeta^* + \text{etc.} - Fv^* \\ &= A + [\alpha\alpha]\xi^* + [\alpha\beta]\eta^* + [\alpha\gamma]\zeta^* + \text{etc.} \\ &\quad - \frac{F}{1+\omega}(F\xi^* + G\eta^* + H\zeta^* + \text{etc.} + K) \end{aligned}$$

From this we therefore conclude that

$$A^* = A - \frac{FK}{1+\omega}$$

which will be the most plausible value of x from *all* the observations;

$$[\alpha\alpha^*] = [\alpha\alpha] - \frac{FF}{1+\omega}$$

and the weight of this determination is therefore

$$= \frac{1}{[\alpha\alpha] - \frac{FF}{1+\omega}}$$

In exactly the same way we find the most plausible value of y, based on *all* the observations,

$$B^* = B - \frac{GK}{1+\omega}$$

and the weight of this determination is

$$= \frac{1}{[\beta\beta] - \frac{GG}{1+\omega}}$$

and so on. Q. E. I.

Let us add a few remarks to this solution.

I. On substituting these new values A^*, B^*, C^* etc., the function v^* acquires the most plausible value

$$K - \frac{K}{1+\omega}(Ff + Gg + Hh + \text{etc.}) = \frac{K}{1+\omega}$$

And since we have indefinitely

$$v^* = \frac{F}{1+\omega}.\xi^* + \frac{G}{1+\omega}.\eta^* + \frac{H}{1+\omega}.\zeta^* + \text{etc.} + \frac{K}{1+\omega}$$

the weight of this determination is found by the principles of art. 29 to be

$$= \frac{1+\omega}{Ff+Gg+Hh+\text{etc.}} = \frac{1}{\omega}+1$$

The same results follow directly by applying the rule given at the end of art. 21; namely, the system of original equations had supplied the determination $v^* = K$ with weight $= \frac{1}{\omega}$, and the new observation then gave another determination, independent of the former, $v^* = 0$, with weight $= 1$; combining them gives the determination $v^* = \frac{K}{1+\omega}$ with weight $= \frac{1}{\omega}+1$.

II. It further follows that, since for $x = A^*, y = B^*, z = C^*$ etc. we must have $\xi^* = 0, \eta^* = 0, \zeta^* = 0$ etc., the same values give

$$\xi = -\frac{fK}{1+\omega}, \quad \eta = -\frac{gK}{1+\omega}, \quad \zeta = -\frac{hK}{1+\omega} \quad \text{etc.}$$

and also, since indefinitely $\Omega = \xi(x-A) + \eta(y-B) + \zeta(z-C) + \text{etc.} + M$,

$$\begin{aligned}\Omega &= \frac{KK}{(1+\omega)^2}(Ff+Gg+Hh+\text{etc.}) + M \\ &= M + \frac{\omega KK}{(1+\omega)^2}\end{aligned}$$

and finally, since indefinitely $\Omega^* = \Omega + v^*v^*$,

$$\Omega^* = M + \frac{\omega KK}{(1+\omega)^2} + \frac{KK}{(1+\omega)^2} = M + \frac{KK}{1+\omega}$$

III. Comparing this with what we taught in art. 30, we observe that the function Ω here attains the smallest value it can take for the specified value of the function $v^* = \frac{K}{1+\omega}$.

36.

For another problem, related to the preceding one, namely

To investigate the changes in the most plausible values of the unknowns arising from a change in the weight of one of the original observations, and the weights of the new determinations

we shall give only the solution here, omitting for brevity the proof, which is easily completed on the model of the preceding article.

Let us suppose that only after completing the calculation we notice that one of the observations has been assigned a weight either too small or too large; e. g. that the first observation, which gave $V = L$, should be assigned,

instead of the weight p used in the calculation, more correctly a weight $= p^*$. There will then be no need to repeat the entire calculation; instead, the corrections can be computed more conveniently by the following formulas.

The corrected most plausible values of the unknowns will be these:

$$x = A - \frac{(p^* - p)\alpha\lambda}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$y = B - \frac{(p^* - p)\beta\lambda}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$z = C - \frac{(p^* - p)\gamma\lambda}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

etc., and the weights of these determinations will be found by dividing unity respectively by

$$[\alpha\alpha] - \frac{(p^* - p)\alpha\alpha}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$[\beta\beta] - \frac{(p^* - p)\beta\beta}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$[\gamma\gamma] - \frac{(p^* - p)\gamma\gamma}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})} \quad \text{etc.}$$

This solution also includes the case where, after completing the calculation, we realize that one of the observations should have been rejected altogether, since this is the same as putting $p^* = 0$; and similarly the value $p^* = \infty$ refers to the case where the equation $V = L$, which was treated as approximate in the calculation, actually has absolute precision.

Moreover, whenever either *several* new equations are added to those on which the calculation was based, or we realize that *several* of them have been assigned incorrect weights, the calculation of the corrections would become too complicated; in such a case it will therefore be better to redo the calculation from the beginning.

37.

In arts. 15. 16 we explained a method for determining the precision of observations approximately[*]. But this method assumes that a sufficiently large number of the errors that actually occurred are known exactly, a condition which, strictly speaking, will very rarely, not to say never, be met. If, indeed, the quantities whose approximate values have become known through observations depend, according to a known law, on one or more unknown quantities, the most plausible values of these unknowns can be found by the method of least squares. The values of the quantities that were the objects of observation, computed from those values, will then be supposed to differ very little from the true values, so that the greater the number of observations, the more justification there will be for taking their differences from the observed values as the true errors of the observations. This practice has been followed by all calculators who have undertaken to estimate the precision of observations a posteriori in particular cases: but it is clearly wrong in theory, and although in many cases it may suffice for practical purposes, in others it can lead to enormous errors. This subject therefore very much deserves a more careful examination.

In this investigation we shall retain the notation used from art. 19 onward. The practice we have described regards the quantities A, B, C etc. as the true values of x, y, z, and consequently $\lambda, \lambda', \lambda''$ etc. as the true values of the functions v, v', v'' etc. If all observations have equal precision and their weight $p = p' = p''$ etc. has been taken as unity, the same quantities, with their signs changed, represent the errors of the observations under that assumption; the rules of art. 15 then give the mean error of the observations m

$$= \sqrt{\frac{\lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.}}{\pi}} = \sqrt{\frac{M}{\pi}}$$

If the observations do not have the same precision, the quantities $-\lambda, -\lambda', -\lambda''$ etc. represent the errors of the observations multiplied by the square roots of their weights, and the rules of art. 16 would lead to the same formula $\sqrt{\frac{M}{\pi}}$, now denoting the mean error of observations assigned weight $= 1$. But clearly an exact calculation would require that, instead of the quantities $\lambda, \lambda', \lambda''$ etc., the values of the functions v, v', v'' etc. obtained from the true values of x, y, z etc. be used; i. e. instead of M, the value of the function Ω corresponding to the true values of x, y, z etc. Although this value cannot be specified, we are nevertheless certain that it is greater than M (which is

[*] The investigation of the same subject which we had presented in an earlier paper (*Bestimmung der Genauigkeit der Beobachtungen. Zeitschrift für Astronomie und verwandte Wissenschaften* Vol. I, p. 185) rested on the same hypothesis concerning the nature of the function expressing the probability of errors on which we had based the method of least squares in Theoria motus corporum coelestium (see art. 9, III).

the smallest possible value), except in the infinitely improbable case where the most plausible values of the unknowns agree exactly with the true values. In general, therefore, we can assert that the usual practice produces a mean error smaller than it should be, or assigns the observations too great a precision. Let us now see what rigorous theory teaches.

38.

First of all it is necessary to investigate how M depends on the true errors of the observations. Let us denote these, as in art. 28, by e, e', e'' etc., and for greater simplicity put

$$e\sqrt{p} = \varepsilon, \quad e'\sqrt{p'} = \varepsilon', \quad e''\sqrt{p''} = \varepsilon'' \quad \text{etc., and also}$$
$$m\sqrt{p} = m'\sqrt{p'} = m''\sqrt{p''} \quad \text{etc.} = \mu$$

Further, let the true values of x, y, z etc. be, respectively, $A - x^0, B - y^0, C - z^0$ etc., and let the corresponding values of ξ, η, ζ etc. be $-\xi^0, -\eta^0, -\zeta^0$ etc. Clearly the same values will give for v, v', v'' etc. the values $-\varepsilon, -\varepsilon', -\varepsilon''$ etc., so that the following holds

$$\xi^0 = a\varepsilon + a'\varepsilon' + a''\varepsilon'' + \text{etc.}$$
$$\eta^0 = b\varepsilon + b'\varepsilon' + b''\varepsilon'' + \text{etc.}$$
$$\zeta^0 = c\varepsilon + c'\varepsilon' + c''\varepsilon'' + \text{etc.}$$

etc., and also

$$x^0 = \alpha\varepsilon + \alpha'\varepsilon' + \alpha''\varepsilon'' + \text{etc.}$$
$$y^0 = \beta\varepsilon + \beta'\varepsilon' + \beta''\varepsilon'' + \text{etc.}$$
$$z^0 = \gamma\varepsilon + \gamma'\varepsilon' + \gamma''\varepsilon'' + \text{etc.}$$

Finally, let us put

$$\Omega^0 = \varepsilon\varepsilon + \varepsilon'\varepsilon' + \varepsilon''\varepsilon'' + \text{etc.}$$

so that Ω^0 is equal to the value of the function Ω corresponding to the true values of x, y, z etc. Since the following holds indefinitely

$$\Omega = M + (x - A)\xi + (y - B)\eta + (z - C)\zeta + \text{etc.}$$

it also follows that

$$M = \Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.}$$

It is therefore clear that M, when expanded, is a homogeneous function of the second order of the errors e, e', e'' etc., which may become larger or smaller according to the different values of the errors. But insofar as the sizes of the errors remain unknown to us, it will be appropriate to consider this function

indefinitely and, above all, to assign its mean value according to the principles of the calculus of probability. We shall find this by writing, in place of the squares $ee, e'e', e''e''$ etc., respectively, $mm, m'm', m''m''$ etc., while omitting the products $ee', ee'', e'e''$ etc. altogether; or, what is the same, by writing in place of each square $\varepsilon\varepsilon, \varepsilon'\varepsilon', \varepsilon''\varepsilon''$ etc. the value $\mu\mu$, and disregarding the products $\varepsilon\varepsilon', \varepsilon\varepsilon'', \varepsilon'\varepsilon''$ etc. entirely. In this way the term Ω^0 clearly gives $\pi\mu\mu$; the term $-x^0\xi^0$ will give

$$-(a\alpha + a'\alpha' + a''\alpha'' + \text{etc.})\mu\mu = -\mu\mu$$

and similarly each of the remaining parts will give $-\mu\mu$, so that the total mean value becomes $= (\pi - \rho)\mu\mu$, where π denotes the number of observations and ρ the number of unknowns. The true value of M may indeed be greater or smaller than the mean, according to the errors chance has produced; but the greater the number of observations, the less significant the discrepancy will be, so that the following can be taken as an approximate value of μ:

$$\sqrt{\frac{M}{\pi - \rho}}$$

Thus the value of μ resulting from the erroneous practice discussed in the preceding article must be increased in the ratio of the quantity $\sqrt{(\pi - \rho)}$ to $\sqrt{\pi}$.

39.

To make clearer how far it is legitimate to equate the value of M arising by chance with its mean, it is still necessary to investigate the mean error to be feared when we put $\frac{M}{\pi-\rho} = \mu\mu$. This mean error is equal to the square root of the mean value of the quantity

$$\left(\frac{\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.} - (\pi-\rho)\mu\mu}{\pi-\rho}\right)^2$$

which we shall express as follows:

$$\left(\frac{\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.}}{\pi-\rho}\right)^2$$
$$- \frac{2\mu\mu}{\pi-\rho}(\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.} - (\pi-\rho)\mu\mu) - \mu^4$$

and since the mean value of the second term is clearly $= 0$, what we have to investigate is the mean value of the function

$$\Psi = (\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.})^2$$

Once this has been found and denoted by N, the required mean error will be

$$= \sqrt{\left(\frac{N}{(\pi-\rho)^2} - \mu^4\right)}$$

The expanded expression Ψ is clearly a homogeneous function either of the errors e, e', e'' etc. or of the quantities $\varepsilon, \varepsilon', \varepsilon''$ etc., and its mean value will be found if

1° the fourth powers e^4, e'^4, e''^4 etc. are replaced by their mean values

2° each product of two squares, such as $eee'e', eee''e'', e'e'e''e''$ etc., is replaced by the product of their mean values, namely $mmm'm', mmm''m'', m'm'm''m''$ etc.

3° all the remaining parts, which will involve either a factor such as e^3e' or one such as $eee'e''$, are omitted altogether.

We shall assume the mean values of the fourth powers e^4, e'^4, e''^4 etc. to be proportional to the fourth powers m^4, m'^4, m''^4 etc. themselves (see art. 16), so that the former are to the latter as ν^4 to μ^4; thus ν^4 denotes the mean value of the fourth powers of observations whose weight is $= 1$. The preceding rules can therefore also be expressed as follows: in place of each fourth power $\varepsilon^4, \varepsilon'^4, \varepsilon''^4$ etc. we must write ν^4, in place of each product of two squares, such as $\varepsilon\varepsilon\varepsilon'\varepsilon', \varepsilon\varepsilon\varepsilon''\varepsilon'', \varepsilon'\varepsilon'\varepsilon''\varepsilon''$ etc., we must write μ^4, and all remaining terms that will involve factors such as $\varepsilon^3\varepsilon'$ or $\varepsilon\varepsilon\varepsilon'\varepsilon''$ or $\varepsilon\varepsilon'\varepsilon''\varepsilon'''$ must be suppressed.

With this properly understood, it will readily be seen that

I. The mean value of the square $\Omega^0\Omega^0$ is $\pi\nu^4 + (\pi\pi - \pi)\mu^4$

II. The mean value of the product $\varepsilon\varepsilon x^0\xi^0$ is $= a\alpha\nu^4 + (a'\alpha' + a''\alpha'' + \text{etc.})\mu^4$, or, since $a\alpha + a'\alpha' + a''\alpha'' + \text{etc.} = 1$,

$$= a\alpha(\nu^4 - \mu^4) + \mu^4$$

And since, similarly, the mean value of the product $\varepsilon'\varepsilon' x^0\xi^0$ is $= a'\alpha'(\nu^4 - \mu^4) + \mu^4$, while the mean value of the product $\varepsilon''\varepsilon'' x^0\xi^0$ is $= a''\alpha''(\nu^4 - \mu^4) + \mu^4$ and so on, it is clear that the mean value of the product $(\varepsilon\varepsilon + \varepsilon'\varepsilon' + \varepsilon''\varepsilon'' + \text{etc.})x^0\xi^0$ or $\Omega^0 x^0\xi^0$ is

$$= \nu^4 - \mu^4 + \pi\mu^4$$

The products $\Omega^0 y^0\eta^0, \Omega^0 z^0\zeta^0$ etc. will have the same mean value. Therefore the mean value of the product $\Omega^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ is

$$= \rho\nu^4 + \rho(\pi - 1)\mu^4$$

III. To keep the remaining expansions from becoming too lengthy, a suitable notation must be introduced. We shall therefore use the symbol Σ in a somewhat wider sense than has been done at various places above: it will denote the sum of the term to which it is prefixed and all similar but non-identical terms arising from it by all permutations of the observations. In this way, e. g., we have $x^0 = \Sigma\alpha\varepsilon$, $x^0x^0 = \Sigma\alpha\alpha\varepsilon\varepsilon + 2\Sigma\alpha\alpha'\varepsilon\varepsilon'$. Collecting the mean value of the product $x^0x^0\xi^0\xi^0$ by parts, we therefore first have the mean value of the product $\alpha\alpha\varepsilon\varepsilon\xi^0\xi^0$

$$\begin{aligned} &= aa\alpha\alpha\nu^4 + \alpha\alpha(a'a' + a''a'' + \text{etc.})\mu^4 \\ &= aa\alpha\alpha(\nu^4 - \mu^4) + \alpha\alpha\mu^4\Sigma aa \end{aligned}$$

Similarly the mean value of the product $\alpha'\alpha'\varepsilon'\varepsilon'\xi^0\xi^0$ is $= a'a'\alpha'\alpha'(\nu^4 - \mu^4) + \alpha'\alpha'\mu^4\Sigma aa$ and so on, and therefore the mean value of the product $\xi^0\xi^0\Sigma\alpha\alpha\varepsilon\varepsilon$ is

$$= (\nu^4 - \mu^4)\Sigma aa\alpha\alpha + \mu^4\Sigma aa.\Sigma\alpha\alpha$$

Further, the mean value of the product $\alpha\alpha'\varepsilon\varepsilon'\xi^0\xi^0$ is $= 2aa'\alpha\alpha'\mu^4$, and similarly the mean value of the product $\alpha\alpha''\varepsilon\varepsilon''\xi^0\xi^0$ is $= 2aa''\alpha\alpha''\mu^4$ etc.; from this it is readily concluded that the mean value of the product $\xi^0\xi^0\Sigma\alpha\alpha'\varepsilon\varepsilon'$ is

$$\begin{aligned} &= 2\mu^4\Sigma a\alpha a'\alpha' = \mu^4((\Sigma a\alpha)^2 - \Sigma aa\alpha\alpha) \\ &= \mu^4(1 - \Sigma aa\alpha\alpha) \end{aligned}$$

Collecting these, we have the mean value of the product $x^0x^0\xi^0\xi^0$

$$= (\nu^4 - 3\mu^4)\Sigma aa\alpha\alpha + 2\mu^4 + \mu^4\Sigma aa.\Sigma\alpha\alpha$$

IV. In a very similar way the mean value of the product $x^0y^0\xi^0\eta^0$ is found to be

$$\begin{aligned} &= \nu^4\Sigma ab\alpha\beta + \mu^4\Sigma a\alpha b'\beta' \\ &\quad + \mu^4\Sigma ab\alpha'\beta' + \mu^4\Sigma a\beta b'\alpha' \end{aligned}$$

But the following holds

$$\begin{aligned} \Sigma a\alpha b'\beta' &= \Sigma a\alpha.\Sigma b\beta - \Sigma a\alpha b\beta \\ \Sigma ab\alpha'\beta' &= \Sigma ab.\Sigma\alpha\beta - \Sigma ab\alpha\beta \\ \Sigma a\beta b'\alpha' &= \Sigma a\beta.\Sigma b\alpha - \Sigma a\beta b\alpha \end{aligned}$$

therefore, because $\Sigma a\alpha = 1, \Sigma b\beta = 1, \Sigma a\beta = 0, \Sigma b\alpha = 0$, that mean value becomes

$$= (\nu^4 - 3\mu^4)\Sigma ab\alpha\beta + \mu^4(1 + \Sigma ab.\Sigma\alpha\beta)$$

V. Since, in exactly the same way, the mean value of the product $x^0z^0\xi^0\zeta^0$ is

$$= (\nu^4 - 3\mu^4)\Sigma ac\alpha\gamma + \mu^4(1 + \Sigma ac.\Sigma\alpha\gamma)$$

and so on, addition gives the mean value of the product $x^0\xi^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ as

$$\begin{aligned} &= (\nu^4 - 3\mu^4)\Sigma(a\alpha(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 1)\mu^4 \\ &\quad + \mu^4(\Sigma aa.\Sigma\alpha\alpha + \Sigma ab.\Sigma\alpha\beta + \Sigma ac.\Sigma\alpha\gamma + \text{etc.}) \\ &= (\nu^4 - 3\mu^4)\Sigma(a\alpha(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 2)\mu^4 \end{aligned}$$

VI. In exactly the same way the mean value of the product $y^0\eta^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ is found to be

$$= (\nu^4 - 3\mu^4)\Sigma(b\beta(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 2)\mu^4$$

then the mean value of the product $z^0\zeta^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ is

$$= (\nu^4 - 3\mu^4)\Sigma(c\gamma(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 2)\mu^4$$

and so on. Addition therefore gives the mean value of the square $(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})^2$

$$= (\nu^4 - 3\mu^4)\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2) + (\rho\rho + 2\rho)\mu^4$$

VII. Finally, collecting everything properly gives

$$\begin{aligned} N &= (\pi - 2\rho)\nu^4 + (\pi\pi - \pi - 2\pi\rho + 4\rho + \rho\rho)\mu^4 \\ &\quad + (\nu^4 - 3\mu^4)\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2) \\ &= (\pi - \rho)(\nu^4 - \mu^4) + (\pi - \rho)^2\mu^4 \\ &\quad - (\nu^4 - 3\mu^4)[\rho - \Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)] \end{aligned}$$

Thus the mean error to be feared in determining $\mu\mu$ by the formula

$$\mu\mu = \frac{M}{\pi - \rho}$$

will be

$$= \sqrt{\left\{ \frac{\nu^4 - \mu^4}{\pi - \rho} - \frac{\nu^4 - 3\mu^4}{(\pi - \rho)^2} . [\rho - \Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)] \right\}}$$

40.

The quantity $\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)$, which enters the expression just found, cannot in general be reduced to a simpler form. Nevertheless two bounds can be assigned between which its value must necessarily lie. *First*, from the relations developed above it is readily proved that

$$\begin{aligned} &(a\alpha + b\beta + c\gamma + \text{etc.})^2 + (a\alpha' + b\beta' + c\gamma' + \text{etc.})^2 \\ &\quad + (a\alpha'' + b\beta'' + c\gamma'' + \text{etc.})^2 + \text{etc.} \\ &= a\alpha + b\beta + c\gamma + \text{etc.} \end{aligned}$$

from which we conclude that $a\alpha + b\beta + c\gamma + \text{etc.}$ is a positive quantity smaller than unity (at least not greater). The same holds for the quantity $a'\alpha' + b'\beta' + c'\gamma' + \text{etc.}$, which is found to be equal to the sum

$$\begin{aligned} &(a'\alpha + b'\beta + c'\gamma + \text{etc.})^2 + (a'\alpha' + b'\beta' + c'\gamma' + \text{etc.})^2 \\ &\quad + (a'\alpha'' + b'\beta'' + c'\gamma'' \text{ etc.})^2 + \text{etc.} \end{aligned}$$

and similarly $a''\alpha'' + b''\beta'' + c''\gamma'' + \text{etc.}$ will be smaller than unity, and so on. Hence $\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)$ is necessarily smaller than π. *Second*, we have $\Sigma(a\alpha + b\beta + c\gamma + \text{etc.}) = \rho$, since $\Sigma a\alpha = 1, \Sigma b\beta = 1, \Sigma c\gamma = 1$ etc.; from this it is readily deduced that the sum of squares $\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)$ is greater than $\frac{\rho\rho}{\pi}$, or at least not smaller. Thus the term

$$\frac{\nu^4 - 3\mu^4}{(\pi - \rho)^2} . [\rho - \Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)]$$

necessarily lies between the bounds $-\frac{\nu^4 - 3\mu^4}{\pi - \rho}$ and $\frac{\nu^4 - 3\mu^4}{\pi - \rho} . \frac{\rho}{\pi}$ or, if we prefer wider ones, between $-\frac{\nu^4 - 3\mu^4}{\pi - \rho}$ and $+\frac{\nu^4 - 3\mu^4}{\pi - \rho}$, and therefore the square of the mean error to be feared in the value of $\mu\mu = \frac{M}{\pi - \rho}$ lies between the bounds $\frac{2\nu^4 - 4\mu^4}{\pi - \rho}$ and $\frac{2\mu^4}{\pi - \rho}$, so that any degree of precision can be attained, provided only that the number of observations is sufficiently large.

It is very remarkable that under the hypothesis (art. 9, III) on which the theory of least squares was formerly based, that term disappears altogether;

and just as, to obtain an approximate value of the mean error of the observations μ, in all cases the sum $\lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.} = M$ must be treated as if it were a sum of $\pi - \rho$ errors arising by chance, so under that hypothesis the precision of this determination itself also becomes equal to that which we found should be assigned to a determination from $\pi - \rho$ true errors in art. 15.
